BIN CHEN

Address: 2233 Tech Drive, Mudd Room 4706, Evanston, IL 60208
Phone: 480-310-8682 ⋄ Email: bin.chen@northwestern.edu ⋄ Website: binchen.ac

EDUCATION

Arizona State University, Tempe, AZ 2014 – 2018
Ph.D. Materials Science and Engineering

Fuzhou University, China 2010 – 2014
B.Eng. Materials Science and Engineering

RESEARCH EXPERIENCE

Northwestern University, Evanston, IL 2025 – present
Director of Research, Paula M. Trienens Institute for Sustainability and Energy

Northwestern University, Evanston, IL 2024 – present
Associate Director, TEAMUP Consortium (Tandems for Efficient and Advanced Modules using Ultra-stable Perovskites)

Northwestern University, Evanston, IL 2022 – present
Research Associate Professor (2024 – present); Research Assistant Professor (2022 – 2024), Department of Chemistry; Department of Electrical & Computer Engineering
Sponsoring Principal Investigator: Prof. Edward H. Sargent
Research: Emerging semiconductors for electronics and optoelectronics

University of Toronto, Canada 2018 – 2022
Postdoctoral Fellow, Department of Electrical & Computer Engineering
Advisor: Prof. Edward H. Sargent
Research: Perovskite-based tandem solar cells and quantum dot infrared photodetectors

Arizona State University, Tempe, AZ 2014 – 2018
Graduate Research Associate, School for Engineering of Matter, Transport & Energy
Advisor: Prof. Sefaattin Tongay
Dissertation: Atomic Scale Characterizations of Two-dimensional Anisotropic Materials and Their Heterostructures
Committee Members: Prof. Sefaattin Tongay, Prof. Mariana Bertoni, Prof. Shery Chang

SELECTED PUBLICATIONS

† equal contribution, * corresponding

26. Yang, Y.; Zhang, X.; Liu, C.; Mikuciunaite, J.; Li, C.; Krisiune, D.; Shibayama, N.; Zhuang, J.; Shi, P.; Shin, D.; Kazianga, U. H.; Huang, Z.; Gilley, I. W.; López-Arteaga, R.; Ginger, D. S.; Getautis, V.; Chen, H.*; Rakstys, K.*; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Improved Crystallization-Front Dynamics Reduce Buried-Interface Defects in Perovskite Solar Cells. *J. Am. Chem. Soc.* **2026**, *148* (41), 44349–44359.

25. Shin, D.†; Kitade, S.†; Duan, L.†; Liu, C.†; Li, C.†; Huang, C.; Kazianga, U. H.; López-Arteaga, R.; Reitz, S. A.; Choi, D.; Yang, Y.; Kim, H. J.; Lee, J.; Beights, J.; Kwon, H. W.; Song, J.-Y.; Liu, Y.; Choi, J.; Mirkin, C. A.; Kanatzidis, M. G.; **Chen, B.***; Sargent, E. H.* Sterically Gated Lewis Acid and Base Pairs Enable Orthogonal Defect Passivation in Perovskite Solar Cells. *Nat. Chem.* **2026**. https://doi.org/10.1038/s41557-026-02261-z.

24. Shi, P.†; Zhuang, J.†; Zhang, D.†; Li, C.†; Zeiske, S.; Hou, J.; Gilley, I. W.; Yavuz, I.; Kazianga, U.; Li, C.; Zhang, Y.; Huang, K.; Itsoponpan, T.; Jiang, X.; Chen, C.-H.; Huang, C.; Yang, Y.; Zhang, X.; Nandi, P.; Reitz, S. A.; Facchetti, A.; White, K. P.; Wyatt, K.; Toney, M. F.; Kanatzidis, M. G.; Marks, T. J.*; Liu, C.*; **Chen, B.***; Sargent, E. H.* Phase-Homogeneous Mixed Halide Perovskites for Stable Tandem Photovoltaics. *Nature* **2026**, *657* (8130), 121–126.

23. Hou, J.; Gilley, I. W.; Wiggins, T. E.; Liu, C.; Yang, Y.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Synthetic Routes to Advancing Perovskite Solar Cells Through Interface Design. *Nat. Synth.* **2026**, *5*, 975–988.

22. Zeiske, S.; Ban, H. W.; Li, X.; Deng, B.; López-Arteaga, R.; Kazianga, U. H.; Han, M. G.; Kim, T.-G.; **Chen, B.***; Sargent, E. H.* Revealing Additional Size-Dependent Defect Suppression Channels Governing Detectivity in InAs Colloidal Quantum Dot Photodiodes. *Nano Lett.* **2025**, *25* (48), 16862–16868.

21. Huang, C.; Yang, Y.; Liu, C.; Chen, H.; Chaudhuri, S.; Jeon, W. C.; Li, M.; Rolston, N.; Bati, A. S. R.; Gilley, I. W.; Kumral, B.; Serles, P.; Filleter, T.; Schatz, G. C.; Kanatzidis, M. G.; **Chen, B.***; Chen, L. X.*; Sargent, E. H.* Electrostatically Enhanced Buried Interface Binding of Self-Assembled Monolayers for Efficient and Stable Inverted Perovskite Solar Cells. *Adv. Mater.* **2025**, *37* (43), e08740.

20. Bati, A. S. R.; Liu, C.; Gilley, I. W.; Musgrave, C. B., III; Maxwell, A.; Steele, J. A.; Yang, Y.; Chen, H.; Wan, H.; Xu, J.; Solano, E.; Zhang, R.; Huang, C.; Rehl, B.; Lempesis, N.; Carnevali, V.; Vezzosi, A.; Zeng, L.; Grater, L.; Li, M.; Rolston, N.; Choi, D.; Sláma, V.; Rothlisberger, U.; Wang, L.; Goddard, W. A., III; Kanatzidis, M. G.; **Chen, B.***; Bakr, O. M.*; Sargent, E. H.* A Chemically Bonded Monolayer Interface Enables Enhanced Thermal Stability and Efficiency in Pb-Sn Perovskite Solar Cells. *Joule* **2025**, *9* (9), 102047.

19. Choi, D.; Shin, D.; Li, C.; Liu, Y.; Bati, A. S. R.; Kachman, D. E.; Yang, Y.; Li, J.; Lee, Y. J.; Li, M.; Penukula, S.; Kim, D. B.; Shin, H.; Chen, C.-H.; Park, S. M.; Liu, C.; Maxwell, A.; Wan, H.; Rolston, N.; Sargent, E. H.*; **Chen, B.*** Carboxyl-Functionalized Perovskite Enables ALD Growth of a Compact and Uniform Ion Migration Barrier. *Joule* **2025**, 9 (3), 101801.

18. Yang, Y.; Chen, H.; Liu, C.; Xu, J.; Huang, C.; Malliakas, C. D.; Wan, H.; Bati, A. S. R.; Wang, Z.; Reynolds, R. P.; Gilley, I. W.; Kitade, S.; Wiggins, T. E.; Zeiske, S.; Suragtkhuu, S.; Batmunkh, M.; Chen, L. X.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Amidination of Ligands for Chemical and Field-Effect Passivation Stabilizes Perovskite Solar Cells. *Science* 2024, *386* (6724), 898–902.

17. Li, C.†; Chen, L.†; Jiang, F.†; Song, Z.†; Wang, X.; Balvanz, A.; Ugur, E.; Liu, Y.; Liu, C.; Maxwell, A.; Chen, H.; Liu, Y.; Wang, Z.; Xia, P.; Li, Y.; Fu, S.; Sun, N.; Grice, C. R.; Wu, X.; Fink, Z.; Hu, Q.; Zeng, L.; Jung, E.; Wang, J.; Park, S. M.; Luo, D.; Chen, C.; Shen, J.; Han, Y.; Perini, C. A. R.; Correa-Baena, J.-P.; Lu, Z.-H.; Russell, T. P.; De Wolf, S.; Kanatzidis, M. G.; Ginger, D. S.; **Chen, B.***; Yan, Y.*; Sargent, E. H.* Diamine Chelates for Increased Stability in Mixed Sn–Pb and All-Perovskite Tandem Solar Cells. *Nat. Energy* 2024, *9* (11), 1388–1396.

16. Chen, H.†; Liu, C.†; Xu, J.†; Maxwell, A.†; Zhou, W.†; Yang, Y.; Zhou, Q.; Bati, A. S. R.; Wan, H.; Wang, Z.; Zeng, L.; Wang, J.; Serles, P.; Liu, Y.; Teale, S.; Liu, Y.; Saidaminov, M.; Hoogland, S.; Filleter, T.; Kanatzidis, M. G.; **Chen, B.***; Ning Z*; Sargent, E. H.* Improved charge extraction in inverted perovskite solar cells with dual-site-binding ligands. *Science* 2024, 384 (6692), 189–193.

15. Xu, J.; Maxwell, A.; Song, Z.; Bati, A. S. R.; Chen, H.; Li, C.; Park, S. M.; Yan, Y.; **Chen, B.***; Sargent, E. H.* The Dynamic Adsorption Affinity of Ligands Is a Surrogate for the Passivation of Surface Defects. *Nat. Commun.* 2024, 15 (1), 2035.

14. Maxwell, A.†; Chen, H.†; Grater, L.; Li, C.; Teale, S.; Wang, J.; Zeng, L.; Wang, Z.; Park, S.

M.; Vafaie, M.; Sidhik, S.; Metcalf, I. W.; Liu, Y.; Mohite, A. D.; **Chen, B.***; Sargent, E. H.* All-Perovskite Tandems Enabled by Surface Anchoring of Long-Chain Amphiphilic Ligands. *ACS Energy Lett.* 2024, *9*, 520–527.

13. Yang, Y.†; Liu, C.†; Ding, Y.†; Ding, B.†; Xu, J.†; Liu, A.; Yu, J.; Grater, L.; Zhu, H.; Hadke, S. S.; Sangwan, V. K.; Bati, A. S. R.; Hu, X.; Li, J.; Park, S. M.; Hersam, M. C.; **Chen, B.***; Nazeeruddin, M. K.*; Kanatzidis, M. G.*; Sargent, E. H.* A Thermotropic Liquid Crystal Enables Efficient and Stable Perovskite Solar Modules. *Nat. Energy* 2024, *9*, 316–323.

12. Liu, C.†; Yang, Y.†; Chen, H.†; Xu, J.†; Liu, A.†; Bati, A. S. R.; Zhu, H.; Grater, L.; Hadke, S. S.; Huang, C.; Sangwan, V. K.; Cai, T.; Shin, D.; Chen, L. X.; Hersam, M. C.; Mirkin, C. A.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Bimolecularly-passivated interface enables efficient and stable inverted perovskite solar cells, *Science* 2023, 382 (6672), 810–815.

11. Liu, Y.†; Zhu, T.†; Grater, L.†; Chen, H.†; Reis, R.; Maxwell, A.; Cheng, M.; Dong, Y.; Teale, S.; Leontowich, A. F. G.; Kim, C.; Chan, P. T.; Wang, M.; Paritmongkol, W.; Gao, Y.; Park, S.; Xu, J.; Khan, J. I.; Laquai, F.; Walker, G. C.; Dravid, V. P.; **Chen, B.***; Sargent, E. H.* A Three-Dimensional Quantum Dot Network Stabilizes Perovskite Solids via Hydrostatic Strain. *Matter* 2024, 7 (1), 107–122.

10. Chen, H.†; Maxwell, A.†; Li, C.†; Teale, S.†; **Chen, B.†**; Zhu, T.; Ugur, E.; Harrison, G.; Grater, L.; Wang, J.; Wang, Z.; Zeng, L.; Park, S. M.; Chen, L.; Serles, P.; Awni, R. A.; Subedi, B.; Zheng, X.; Xiao, C.; Podraza, N. J.; Filleter, T.; Liu, C.; Yang, Y.; Luther, J. M.; De Wolf, S.; Kanatzidis, M. G.; Yan, Y.; Sargent, E. H. Regulating Surface Potential Maximizes Voltage in All-Perovskite Tandems. *Nature* 2023, 613 (7945), 676–681.

9. Chen, H.†; Teale, S.†; **Chen, B.†**; Hou, Y.†; Grater, L.; Zhu, T.; Bertens, K.; Park, S. M.; Atapattu, H. R.; Gao, Y.; Wei, M.; Johnston, A. K.; Zhou, Q.; Xu, K.; Yu, D.; Han, C.; Cui, T.; Jung, E. H.; Zhou, C.; Zhou, W.; Proppe, A. H.; Hoogland, S.; Laquai, F.; Filleter, T.; Graham, K. R.; Ning, Z.; Sargent, E. H. Quantum-Size-Tuned Heterostructures Enable Efficient and Stable Inverted Perovskite Solar Cells. *Nat. Photonics* 2022, 16 (5), 352–358.

8. **Chen, B.**; Sargent, E. H. What Does Net Zero by 2050 Mean to the Solar Energy Materials Researcher? *Matter* 2022, 5 (5), 1322–1325.

7. **Chen, B.**; Chen, H.; Hou, Y.; Xu, J.; Teale, S.; Bertens, K.; Chen, H.; Proppe, A.; Zhou, Q.; Yu, D.; Xu, K.; Vafaie, M.; Liu, Y.; Dong, Y.; Jung, E. H.; Zheng, C.; Zhu, T.; Ning, Z.; Sargent, E. H. Passivation of the Buried Interface via Preferential Crystallization of 2D Perovskite on Metal Oxide Transport Layers. *Adv. Mater.* 2021, e2103394.

6. Fang, Z.†; Wang, L.†; Mu, X.†; **Chen, B.†**; Xiong, Q.; Wang, W. D.; Ding, J.; Gao, P.; Wu, Y.; Cao, J. Grain Boundary Engineering with Self-Assembled Porphyrin Supramolecules for Highly Efficient Large-Area Perovskite Photovoltaics. *J. Am. Chem. Soc.* 2021.

5. Jung, E. H.†; **Chen, B.†**; Bertens, K.; Vafaie, M.; Teale, S.; Proppe, A.; Hou, Y.; Zhu, T.; Zheng, C.; Sargent, E. H. Bifunctional Surface Engineering on SnO2 Reduces Energy Loss in Perovskite Solar Cells. *ACS Energy Lett.* 2020, 5 (9), 2796–2801.

4. **Chen, B.**; Baek, S.-W.; Hou, Y.; Aydin, E.; De Bastiani, M.; Scheffel, B.; Proppe, A.; Huang, Z.; Wei, M.; Wang, Y.-K.; Jung, E.-H.; Allen, T. G.; Van Kerschaver, E.; García de Arquer, F. P.; Saidaminov, M. I.; Hoogland, S.; De Wolf, S.; Sargent, E. H. Enhanced Optical Path and Electron Diffusion Length Enable High-Efficiency Perovskite Tandems. *Nat. Commun.* 2020, 11 (1), 1257.

3. Manekkathodi, A.†; **Chen, B.†**; Kim, J.; Baek, S.-W.; Scheffel, B.; Hou, Y.; Ouellette, O.; Saidaminov, M. I.; Voznyy, O.; Madhavan, V. E. Solution-Processed Perovskite-Colloidal Quantum Dot Tandem Solar Cells for Photon Collection beyond 1000 Nm. *Journal of Materials Chemistry A* 2019, 7 (45), 26020–26028.

2. **Chen, B.**; Wu, K.; Suslu, A.; Yang, S.; Cai, H.; Yano, A.; Soignard, E.; Aoki, T.; March, K.; Shen, Y. Controlling Structural Anisotropy of Anisotropic 2D Layers in Pseudo-1D/2D Material Heterojunctions. *Adv. Mater.* 2017, 29 (34), 1701201.

1. **Chen, B.**; Sahin, H.; Suslu, A.; Ding, L.; Bertoni, M. I.; Peeters, F.; Tongay, S. Environmental Changes in MoTe2 Excitonic Dynamics by Defects-Activated Molecular Interaction. *ACS nano* 2015, 9 (5), 5326–5332.

FUNDING SUPPORT

Pending Research

8. NSF 26-512
   U.S. National Science Foundation
   LEAP-PV: Layer-resolved, Evidence-traced, AI-ready Platform for Photovoltaics
   Role: Co-Principal Investigator

7. FY2026 Space Photovoltaics R&D Program (PARSECS)
   U.S. Department of Energy
   $1,500,000 (Northwestern share $1,020,000)
   PERSIST: Perovskite Passivation Engineering for Radiation- and Space-Induced Stress Tolerance
   Role: Principal Investigator

6. FY2026 Space Photovoltaics R&D Program (PARSECS)
   U.S. Department of Energy
   $1,500,000 (Northwestern share $1,050,000)
   VERTEX: Voltage-Enhanced, Robust All-Perovskite Triple-Junctions for EXtreme Specific Power
   Role: Co-Principal Investigator

5. FY2026 Space Photovoltaics R&D Program (PARSECS)
   U.S. Department of Energy (subaward from Verde Technologies)
   $400,000 (Northwestern share)
   All-Perovskite Tandems on Flexible Substrates for Space Applications
   Role: Co-Principal Investigator

4. DOI-BAA-SOLSTICE-FY25-01
   Intelligence Advanced Research Projects Activity (IARPA)
   $600,000 (Northwestern share)
   Superior Options for Long-life Solar Technologies with Impressive Conversion Efficiencies (SOLSTICE)
   Role: Co-Principal Investigator

3. HPC4EI Spring 2026
   U.S. Department of Energy, High Performance Computing for Energy Innovation Program
   $200,000 (Northwestern share)
   Physics-Aware Discovery of Stable Perovskite–Molecular Interfaces for Durable U.S.-Manufactured Tandem Photovoltaic Modules
   Role: Principal Investigator

2. NSF 23-612
   U.S. National Science Foundation
   $600,000
   Synthesis and surface chemistry of InAs colloidal quantum dots toward efficient short-wave infrared detection
   Role: Co-Principal Investigator

1. DE-FOA-0003337, Photovoltaics Research and Development (PVRD) 2024
   U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy

$1,500,000
Efficient and stable near-infrared organic photovoltaics for hybrid organic tandem solar cells
Role: Co-Principal Investigator

Current Research

2. DE-EE0010502
U.S. Department of Energy, Office of Critical Minerals and Energy Innovation
$9,000,000
09/2023 – 10/2025; relaunched 07/2026 – 09/2027
TEAMUP: Tandems for Efficient and Advanced Modules using Ultrastable Perovskites
Role: Senior Personnel; Associate Director (2024 – present)

1. CRG2024
King Abdullah University of Science and Technology
$180,000 (Northwestern share)
2025 – 2028
PRENNIAL
Role: Co-Principal Investigator

Completed Research

7. DE-FOA-0003058
U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy
$500,000 (Northwestern share)
Awarded 09/2024 – 08/2026; terminated by the sponsor 10/2025 in the government-wide DOE project terminations
STACKED: Stability and Characterization of Hole-Transporting Layers Key to Enabling Outdoor Durability
Role: Co-Principal Investigator

6. HR001122S0044-SNAP-FP-009
U.S. Department of Defense, Defense Advanced Research Projects Agency
$10,000,000
Awarded 08/2023 – 07/2027; closed out 11/2025
SYNCED: Interfacing Synthetic Biology with Electrochemical Detectors for Smart Non-Invasive Assays of Physiology
Role: Senior Personnel

5. OSR-5624
King Abdullah University of Science and Technology
$350,000
02/2024 – 01/2026
All-Perovskite Tandem Solar Cells
Role: Co-Principal Investigator

4. Seed Funding Initiative
Center for Engineering Sustainability and Resilience (CESR), Northwestern University
$80,000
02/2024 – 08/2025
Introducing AC Photo-Hall Method: Separating Electron/Hole Mobilities in Perovskite Photovoltaics
Role: Co-Principal Investigator

3. Translation and Incubation Fund
Paula M. Trienens Institute for Sustainability and Energy, Northwestern University
$100,000

01/2024 – 12/2024
Stable perovskite solar cells with cost-effective bilayer metal oxides as electron transport layers
Role: Principal Investigator

2. N00014-20-1-2572
U.S. Department of the Navy, Office of Naval Research
$480,000
08/2020 – 07/2023
Wide-bandgap perovskites for efficient, stable tandems
Role: Senior Personnel

1. OSR-2020-CRG9-4350.2
King Abdullah University of Science and Technology, Office of Sponsored Research
$600,000
04/2021 – 03/2024
SOLSTICE: Solar-driven Circular Carbon Enabled by Perovskite/Perovskite/Si Triple-Junction Tandems
Role: Senior Personnel

PATENTS

3. US Patent Application: Passivating Perovskite Optoelectronic Devices (Application No. 19/375,634, 2026)

2. US and Canadian Patent: A Surface Treatment Method to Passivate Inverted Structure Perovskite Solar Cells

1. US Patent Application: Perovskite Solar Cells With Dual Site Binding Ligands

INVITED TALKS

22. MRS Fall Meeting, December 2026

21. IPS-25, the 25th International Conference on Photochemical Conversion and Storage of Solar Energy, Seoul, July 2026

20. 6th tandemPV Workshop, Berlin, June 2026

19. MRS Spring Meeting, April 2026

18. nanoGe – MATSUS Fall 2025, October 2025

17. Lawrence Symposium on Epitaxy, Arizona State University, October 2025

16. The University of Toledo, October 2025

15. MRS Spring Meeting, April 2025

14. Ludwig Maximilian University of Munich, March 2025

13. Nanyang Technological University, March 2025

12. National University of Singapore, February 2025

11. 4th tandemPV Workshop, June 2024

10. Physics Seminar Series, UC Merced, October 2023

9. 2nd Northwestern/Münster Symposium on Smart Materials, August 2023

8. Homeland Defense & Security Information Analysis Center, April 2023

7. APS March Meeting, March 2023

6. EcoMat Webinar: Perovskite Materials for Photovoltaic and Optoelectronic Applications (virtual), January 2023
5. Lawrence Symposium on Epitaxy, Arizona State University, January 2023
4. Zhejiang University, June 2022
3. KAUST Research Conference: 2022 Accelerating Solar Energy Research towards meeting Vision 2030 Goals, May 2022
2. ICFO – UofT – Stanford International School on the Frontiers of Light, October 2021
1. MRS Spring Meeting (virtual), April 2021

CONFERENCE ORGANIZATION & SERVICE

7. Organizer, nanoGe MATSUS Spring Meeting Symposium: Multilayer Perovskite Optoelectronics, Barcelona, Spring 2027
6. Organizer, ACS Fall Meeting Symposium: Materials Chemistry for Space Power Technologies, August 2026
5. Organizer, Northwestern–HZB–CU Boulder Tandem PV Workshop, May 2026
4. Organizer, ACS Spring Meeting Symposium: Organic, Perovskite, and Hybrid Solar Cells, March 2026
3. Organizer, MRS Spring Meeting Symposium, April 2025
2. Session chair, MRS Fall Meeting, November 2023
1. Organizer, ACS Fall Meeting Symposium: Organic, Perovskite and Hybrid Solar (raised $5,000 sponsorship for the symposium), August 2023

SELECTED ORAL & POSTER PRESENTATIONS

7. Oral, MRS Fall Meeting, December 2026
6. Oral, MRS Fall Meeting, December 2024
5. Oral, ACS Fall Meeting, August 2024
4. Oral, MRS Fall Meeting, November 2023
3. Oral, IEEE PVSC 50th, June 2023
2. Poster, MRS Spring Meeting, March 2017
1. Poster, MRS Spring Meeting, March 2016

TEACHING AND MENTORING

Faculty mentor, Kellogg-Q Entrepreneurial Residency 2024
Kellogg School of Management and the Querrey InQbation Lab, Northwestern University. Hosted MBA students embedded in our laboratory through this independent-study residency (Spring and Summer 2024), mentoring them one-on-one on the commercialization pathway for the perovskite photovoltaic technologies developed in the group; the residents presented their analyses to the full residency cohort and faculty.

Independent-study supervisor, ISP 398 Spring 2024
Weinberg College of Arts and Sciences, Northwestern University. Supervised an undergraduate independent study on solar cell physics and optoelectronic materials: designed the syllabus and reading plan, and guided the student from fundamentals to current research literature.

Independent-study supervisor, Mechanical Engineering 399 Fall 2024
McCormick School of Engineering, Northwestern University. Supervised an undergraduate independent project on the life-cycle assessment of triple-junction perovskite solar cells, guiding the student through the sustainability analysis of an emerging photovoltaic technology.

Postdoctoral and Ph.D. mentoring, Northwestern University 2022 – present
I direct the optoelectronics group, currently mentoring 16 postdoctoral fellows and research staff, and 8 Ph.D. students.
Current postdoctoral fellows and research staff: Linrui Duan, Hangyu Gu, Jin Hou, Xin Jiang, Saikiran Khamgaonkar, Chun-Hsiao Kuan, Chu Li, Shunde Li, Zheng Liang, Pengju Shi, Mengru Wang, Yurui Wang, Lewei Zeng, Xianfu Zhang, Yu Zhang, Qilin Zhou.
Current Ph.D. students: Isaiah Gilley, Chuying Huang, Kefu Huang, Ubaid Kazianga, Samantha Reitz, Matthias Schmidt, Taylor Wiggins, Jiale Zhuang.
Alumni placements: Dr. Somin Park (faculty, National University of Singapore); Dr. Cheng Liu (faculty, Shanghai Jiao Tong University); Dr. Yi Yang (faculty, Shanghai Jiao Tong University); Dr. Hao Chen (faculty, Shanghai Jiao Tong University); Dr. Chongwen Li (faculty, Southeast University); Dr. Guixiang Li (faculty, Southeast University); Dr. Yoon Jung Lee (faculty, Kookmin University); Dr. Deokjae Choi (Senior Researcher, Korea Institute of Science and Technology); Dr. Yuan Liu (Kadanoff-Rice Postdoctoral Fellow, University of Chicago); Dr. Donghoon Shin (Sejong Science Fellow, KAIST); Dr. Stefan Zeiske (promoted to Research Assistant Professor, Northwestern University).

Perovskite Photovoltaic Research Group Leader, University of Toronto 2019 – 2022
I manage a team of over 15 members focusing on perovskite photovoltaics research. My responsibilities include mentoring on specific research projects, conceptualizing manuscripts, writing grant proposals, and preparing grant reports.

Lecturer (simulated), University of Toronto 2019
At the Teaching in Higher Education course at the University of Toronto, I developed my syllabus on "Two-dimensional semiconductor materials and systems" and taught in simulated classes.

ACADEMIC AND SOCIAL SERVICE

Journal advisory board 2025 – present
EES Solar

Journal reviewer 2016 – present
Science, Nature, Nature Energy, Nature Photonics, Nature Sustainability, Chemical Reviews, Journal of the American Chemical Society, Joule, Angewandte Chemie; Nature Communications, Advanced Materials, Matter, Energy & Environmental Science, Advanced Energy Materials, ACS Energy Letters, ACS Nano, ACS Photonics, Advanced Science, Chemical Science, EES Solar, Journal of Materials Chemistry C, ACS Applied Materials & Interfaces, ACS Applied Energy Materials; The Journal of Physical Chemistry, The Journal of Physical Chemistry Letters, Journal of Applied Physics, Progress in photovoltaics, Journal of Photovoltaics, Journal of Physics D: Applied Physics, Journal of Physics: Condensed Matter, 2D Materials, Nanotechnology, Nano Energy

Proposal reviewer 2023 – present
U.S. National Science Foundation
Natural Sciences and Engineering Research Council of Canada
ACS Petroleum Research Fund
National Research Foundation of Singapore
Ministry of Education, Singapore (Academic Research Fund Tier 3)

HONORS AND AWARDS

3. Highly Cited Researcher in the field of Cross-Field - 2023 (Clarivate)

2. Highly Cited Researcher in the field of Cross-Field - 2024 (Clarivate)

1. Highly Cited Researcher in the field of Cross-Field - 2025 (Clarivate)

ALL PUBLICATIONS

147. Yang, Y.; Zhang, X.; Liu, C.; Mikuciunaite, J.; Li, C.; Krisiune, D.; Shibayama, N.; Zhuang, J.; Shi, P.; Shin, D.; Kazianga, U. H.; Huang, Z.; Gilley, I. W.; López-Arteaga, R.; Ginger, D. S.; Getautis, V.; Chen, H.*; Rakstys, K.*; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Improved Crystallization-Front Dynamics Reduce Buried-Interface Defects in Perovskite Solar Cells. *J. Am. Chem. Soc.* **2026**, *148* (41), 44349–44359.

146. Shin, D.; Zeiske, S.; Kim, J.; Kazianga, U. H.; Kim, H.; Jeong, H.; Song, J. H.; **Chen, B.**; Sargent, E. H.; Jeong, S. Overcoming Field-Assisted Leakage and Mobility Deficits in InAs Colloidal Quantum Dot SWIR Photodetectors. *Nano Lett.* **2026**, *26* (39), 13590–13599.

145. Wiggins, T. E.; Bayikadi, K. S.; Kepenekian, M.; Liu, Z.; Peters, J. A.; Tailor, N. K.; Odusote, O. A.; Reynolds, A. P.; **Chen, B.**; Sargent, E. H.; Kanatzidis, M. G. Metformin Spacer Conformation Is a Structure-Directing Element in a 2D Perovskite with Strong X-ray Response. *J. Am. Chem. Soc.* **2026**, *148* (39), 42312–42328.

144. Shin, D.†; Kitade, S.†; Duan, L.†; Liu, C.†; Li, C.†; Huang, C.; Kazianga, U. H.; López-Arteaga, R.; Reitz, S. A.; Choi, D.; Yang, Y.; Kim, H. J.; Lee, J.; Beights, J.; Kwon, H. W.; Song, J.-Y.; Liu, Y.; Choi, J.; Mirkin, C. A.; Kanatzidis, M. G.; **Chen, B.***; Sargent, E. H.* Sterically Gated Lewis Acid and Base Pairs Enable Orthogonal Defect Passivation in Perovskite Solar Cells. *Nat. Chem.* **2026**. https://doi.org/10.1038/s41557-026-02261-z.

143. Shi, P.†; Zhuang, J.†; Zhang, D.†; Li, C.†; Zeiske, S.; Hou, J.; Gilley, I. W.; Yavuz, I.; Kazianga, U.; Li, C.; Zhang, Y.; Huang, K.; Itsoponpan, T.; Jiang, X.; Chen, C.-H.; Huang, C.; Yang, Y.; Zhang, X.; Nandi, P.; Reitz, S. A.; Facchetti, A.; White, K. P.; Wyatt, K.; Toney, M. F.; Kanatzidis, M. G.; Marks, T. J.*; Liu, C.*; **Chen, B.***; Sargent, E. H.* Phase-Homogeneous Mixed Halide Perovskites for Stable Tandem Photovoltaics. *Nature* **2026**, *657* (8130), 121–126.

142. Ban, H. W.; Li, X.; Zeiske, S.; Steele, J. A.; López-Arteaga, R. E.; Lu, H.; Zou, Y.; Han, M. G.; Kim, T.-G.; Liu, C.; **Chen, B.**; Sargent, E. H. Short-Chain Acids Sustain InAs Colloidal Quantum Dot Growth during Synthesis, Extending Spectral Response into the Deep Short-Wave Infrared. *J. Am. Chem. Soc.* **2026**, *148* (21), 21885–21894.

141. Artuk, K.; Türkay, D.; Kuba, A.; Riemelmoser, S.; Steele, J. A.; Hurni, J.; Spitznagel, J.; Quest, H.; De Bastiani, M.; Zhao, J.; Diekmann, J.; Ongaro, C.; Othman, M.; Heydarian, M.; Fischer, O.; Lai, H.; Austin, J. S.; Zeiske, S.; López-Arteaga, R.; Liu, C.; Mensi, M. D.; Castro-Mendez, A.-F.; Li, M.; Gries, T. W.; Hill, S.; Saenz, F.; Champault, L.; Can, H. A.; Golobostanfard, M. R.; Desai, U.; Remondeau, P.; Solano, E.; Portale, G.; Faes, A.; Lang, F.; Musiienko, A.; Rolston, N.; Fu, F.; Schubert, M. C.; Schindler, F.; **Chen, B.**; Pasquarello, A.; Sargent, E. H.; Hessler-Wyser, A.; Jeangros, Q.; Ballif, C.; Wolff, C. M. Triple-Junction Solar Cells with Improved Carrier and Photon Management. *Nature* **2026**, *653* (8113), 90–97.

140. Hou, J.; Fletcher, J.; Hall, S. J.; Zhang, H.; Zacharias, M.; Volonakis, G.; Welton, C.; Zeiske, S.; Gilley, I. W.; Shin, D.; Mandani, F.; Metcalf, I.; Sun, S.; Zhang, B.; Guo, Y.; **Chen, B.**; Reddy, G. N. M.; Katan, C.; Even, J.; Sfeir, M. Y.; Kanatzidis, M. G.; Mohite, A. D. Exciton Diffusion beyond 2 m Enabled by Maximum Symmetry in Two-Dimensional Perovskites. *Nat. Synth.* **2026**, *5* (8), 1254–1268.

139. Choi, D.; Kim, J.; Jaffer, S.; **Chen, B.**; Xie, K.; Sargent, E. H. Translating Insights from Progress in Photovoltaics to Accelerate Industrial-Scale CO2 Electroreduction. *Nat. Energy* **2026**, *11* (2), 185–193.

138. Hou, J.; Gilley, I. W.; Wiggins, T. E.; Liu, C.; Yang, Y.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Synthetic Routes to Advancing Perovskite Solar Cells Through Interface Design. *Nat. Synth.* **2026**, *5*, 975–988.

137. Zeiske, S.; Ban, H. W.; Li, X.; Deng, B.; López-Arteaga, R.; Kazianga, U. H.; Han, M. G.; Kim, T.-G.; **Chen, B.***; Sargent, E. H.* Revealing Additional Size-Dependent Defect Suppression Channels Governing Detectivity in InAs Colloidal Quantum Dot Photodiodes. *Nano Lett.* **2025**, *25* (48), 16862–16868.

136. Huang, C.; Yang, Y.; Liu, C.; Chen, H.; Chaudhuri, S.; Jeon, W. C.; Li, M.; Rolston, N.; Bati, A. S. R.; Gilley, I. W.; Kumral, B.; Serles, P.; Filleter, T.; Schatz, G. C.; Kanatzidis, M. G.; **Chen, B.***; Chen, L. X.*; Sargent, E. H.* Electrostatically Enhanced Buried Interface Binding of Self-Assembled Monolayers for Efficient and Stable Inverted Perovskite Solar Cells. *Adv. Mater.* **2025**, *37* (43), e08740.

135. Bati, A. S. R.; Liu, C.; Gilley, I. W.; Musgrave, C. B., III; Maxwell, A.; Steele, J. A.; Yang, Y.; Chen, H.; Wan, H.; Xu, J.; Solano, E.; Zhang, R.; Huang, C.; Rehl, B.; Lempesis, N.; Carnevali, V.; Vezzosi, A.; Zeng, L.; Grater, L.; Li, M.; Rolston, N.; Choi, D.; Sláma, V.; Rothlisberger, U.; Wang, L.; Goddard, W. A., III; Kanatzidis, M. G.; **Chen, B.***; Bakr, O. M.*; Sargent, E. H.* A Chemically Bonded Monolayer Interface Enables Enhanced Thermal Stability and Efficiency in Pb-Sn Perovskite Solar Cells. *Joule* **2025**, *9* (9), 102047.

134. Hidayatova, L.; Mi, C.; Akhmedov, N. G.; Liu, Y.; Shafiq, A. K.; Afshari, H.; Mohamed-Raseek, N.; Popy, D. A.; Xiang, S.; Chen, Y.-C.; Saparov, B.; Peters, J. W.; Talapin, D. V.; **Chen, B.**; Furis, M.; Glaser, E. R.; Dong, Y. Efficient Mn^{2+} Doping in Non-Stoichiometric Cesium Lead Bromide Perovskite Quantum Dots. *J. Am. Chem. Soc.* **2025**, *147* (38), 35069–35080.

133. Fu, S.; Sun, N.; Chen, H.; Liu, C.; Wang, X.; Li, Y.; Abudulimu, A.; Xu, Y.; Ramakrishnan, S.; Li, C.; Yang, Y.; Wan, H.; Huang, Z.; Xian, Y.; Yin, Y.; Zhu, T.; Chen, H.; Rahimi, A.; Saeed, M. M.; Zhang, Y.; Yu, Q.; Ginger, D. S.; Ellingson, R. J.; **Chen, B.**; Song, Z.; Kanatzidis, M. G.; Sargent, E. H.; Yan, Y. On-Demand Formation of Lewis Bases for Efficient and Stable Perovskite Solar Cells. *Nat. Nanotechnol.* **2025**, *20* (6), 772–778.

132. Choi, D.; Shin, D.; Li, C.; Liu, Y.; Bati, A. S. R.; Kachman, D. E.; Yang, Y.; Li, J.; Lee, Y. J.; Li, M.; Penukula, S.; Kim, D. B.; Shin, H.; Chen, C.-H.; Park, S. M.; Liu, C.; Maxwell, A.; Wan, H.; Rolston, N.; Sargent, E. H.*; **Chen, B.*** Carboxyl-Functionalized Perovskite Enables ALD Growth of a Compact and Uniform Ion Migration Barrier. *Joule* **2025**, *9* (3), 101801.

131. Gilley, I. W.; Kwon, H. W.; Liu, C.; Yang, Y.; Huang, C.; Wan, H.; Bati, A. S. R.; Oriel, E. H.; Kepenekian, M.; Vishal, B.; Zeiske, S.; Bayikadi, K. S.; Wiggins, T. E.; Vasileiadou, E. S.; **Chen, B.**; Schaller, R. D.; Even, J.; De Wolf, S.; Sargent, E. H.; Kanatzidis, M. G. Combining Organic Cations of Different Sizes Grants Improved Control over Perovskitoid Dimensionality and Bandgap. *J. Am. Chem. Soc.* **2025**, *147* (9), 7777–7787.

130. Aryal, S.; Puthirath, A. B.; Jones, B.; Bati, A. S. R.; **Chen, B.**; Mather, T.; Ajayan, P. M.; Sargent, E. H.; Kaul, A. B. Solution-Processed Tungsten Diselenide as an Inorganic Hole Transport Material for Moisture-Stable Perovskite Solar Cells in the n-i-p Architecture. *Sol. Energy Mater. Sol. Cells* **2025**, *282* (113313), 113313.

129. Wiggins, T. E.; Bati, A. S. R.; Kepenekian, M.; Reynolds, R. P.; Zeiske, S.; Pournara, A. D.; Liu, C.; Yang, Y.; Zhang, X.; Gilley, I. W.; **Chen, B.**; Even, J.; Sargent, E. H.; Kanatzidis, M. G. Pb:Sn Ratio-Driven Polytypism and Band Modulation in Photoresponsive Hexagonal Perovskitoids. *J. Am. Chem. Soc.* **2025**, *147* (24), 21014–21031.

128. Chen, J.-K.; Zhou, Y.; Zhang, B.-B.; Kikkawa, J.; Yin, J.; Shirahata, N.; **Chen, B.**; Sargent, E. H.; Sun, H.-T. Computationally Guided Defect-Suppressing Synthesis of Luminescent Tin Halide Perovskite Nanocrystals. *Nat. Synth.* **2025**, *4* (9), 1095–1105.

127. Liu, C.; Yang, Y.; Fletcher, J. D.; Liu, A.; Gilley, I. W.; Musgrave, C. B., III; Wang, Z.; Zhu, H.; Chen, H.; Reynolds, R. P.; Ding, B.; Ding, Y.; Zhang, X.; Skackauskaite, R.; Wan, H.; Zeng, L.; Bati, A. S. R.; Shibayama, N.; Getautis, V.; **Chen, B.**; Rakstys, K.; Dyson, P. J.; Kanatzidis, M.

G.; Sargent, E. H.; Nazeeruddin, M. K. Cation Interdiffusion Control for 2D/3D Heterostructure Formation and Stabilization in Inorganic Perovskite Solar Modules. *Nat. Energy* **2025**, *10* (8), 981–990.

126. Yang, Y.; Chen, H.; Liu, C.; Xu, J.; Huang, C.; Malliakas, C. D.; Wan, H.; Bati, A. S. R.; Wang, Z.; Reynolds, R. P.; Gilley, I. W.; Kitade, S.; Wiggins, T. E.; Zeiske, S.; Suragtkhuu, S.; Batmunkh, M.; Chen, L. X.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Amidination of Ligands for Chemical and Field-Effect Passivation Stabilizes Perovskite Solar Cells. *Science* **2024**, *386* (6724), 898–902.

125. Shen, Y.; Li, C.; Liu, C.; Reitz, S. A.; **Chen, B.**; Sargent, E. H. The Impact of Interface and Heterostructure on the Stability of Perovskite-Based Solar Cells. *Appl. Phys. Rev.* **2024**, *11* (4), 041306.

124. Zhu, T.; Grater, L.; Teale, S.; Vasileiadou, E. S.; Sharir-Smith, J.; **Chen, B.**; Kanatzidis, M. G.; Sargent, E. H. Coupling Photogeneration with Thermodynamic Modeling of Light-Induced Alloy Segregation Enables the Identification of Stabilizing Dopants. Chem. Mater. 2024.

123. Liu, C.; Yang, Y.; Chen, H.; Spanopoulos, I.; Bati, A. S. R.; Gilley, I. W.; Chen, J.; Maxwell, A.; Vishal, B.; Reynolds, R. P.; Wiggins, T. E.; Wang, Z.; Huang, C.; Fletcher, J.; Liu, Y.; Chen, L. X.; De Wolf, S.; **Chen, B.**; Zheng, D.; Marks, T. J.; Facchetti, A.; Sargent, E. H.; Kanatzidis, M. G. Two-Dimensional Perovskitoids Enhance Stability in Perovskite Solar Cells. Nature 2024, 633 (8029), 359–364.

122. Li, S.; Jiang, Y.; Xu, J.; Wang, D.; Ding, Z.; Zhu, T.; **Chen, B.**; Yang, Y.; Wei, M.; Guo, R.; Hou, Y.; Chen, Y.; Sun, C.; Wei, K.; Qaid, S. M. H.; Lu, H.; Tan, H.; Di, D.; Chen, J.; Grätzel, M.; Sargent, E. H.; Yuan, M. High-Efficiency and Thermally Stable FACsPbI3 Perovskite Photovoltaics. *Nature* **2024**, *635* (8037), 82–88.

121. Li, C.†; Chen, L.†; Jiang, F.†; Song, Z.†; Wang, X.; Balvanz, A.; Ugur, E.; Liu, Y.; Liu, C.; Maxwell, A.; Chen, H.; Liu, Y.; Wang, Z.; Xia, P.; Li, Y.; Fu, S.; Sun, N.; Grice, C. R.; Wu, X.; Fink, Z.; Hu, Q.; Zeng, L.; Jung, E.; Wang, J.; Park, S. M.; Luo, D.; Chen, C.; Shen, J.; Han, Y.; Perini, C. A. R.; Correa-Baena, J.-P.; Lu, Z.-H.; Russell, T. P.; De Wolf, S.; Kanatzidis, M. G.; Ginger, D. S.; **Chen, B.***; Yan, Y.*; Sargent, E. H.* Diamine Chelates for Increased Stability in Mixed Sn–Pb and All-Perovskite Tandem Solar Cells. *Nat. Energy* **2024**, *9* (11), 1388–1396.

120. Teale, S.; Degani, M.; **Chen, B.**; Sargent, E. H.; Grancini, G. Molecular Cation and Low-Dimensional Perovskite Surface Passivation in Perovskite Solar Cells. Nat. Energy 2024, 9 (7), 779–792.

119. Khan, J. I.; Yang, Y.; Palmer, J. R.; Tyndall, S. B.; Chaudhuri, S.; Liu, C.; Grater, L.; North, J. D.; **Chen, B.**; Young, R. M.; Schatz, G. C.; Wasielewski, M. R.; Kanatzidis, M. G.; Swearer, D. F.; Sargent, E. H. Evaluation of Interfacial Photophysical Processes by Time-Resolved Optical Spectroscopy in Perovskite Solar Cells. Matter 2024, 7 (7), 2536–2550.

118. Fu, S.; Sun, N.; Xian, Y.; Chen, L.; Li, Y.; Li, C.; Abudulimu, A.; Kaluarachchi, P. N.; Huang, Z.; Wang, X.; Dolia, K.; Ginger, D. S.; Heben, M. J.; Ellingson, R. J.; **Chen, B.**; Sargent, E. H.; Song, Z.; Yan, Y. Suppressed Deprotonation Enables a Durable Buried Interface in Tin-Lead Perovskite for All-Perovskite Tandem Solar Cells. Joule 2024. https://doi.org/10.1016/j.joule.2024.05.007.

117. Chen, H.; Liu, C.; Xu, J.; Maxwell, A.; Zhou, W.; Yang, Y.; Zhou, Q.; Bati, A. S. R.; Wan, H.; Wang, Z.; Zeng, L.; Wang, J.; Serles, P.; Liu, Y.; Teale, S.; Liu, Y.; Saidaminov, M. I.; Li, M.; Rolston, N.; Hoogland, S.; Filleter, T.; Kanatzidis, M. G.; **Chen, B.***; Ning, Z.*; Sargent, E. H.* Improved Charge Extraction in Inverted Perovskite Solar Cells with Dual-Site-Binding Ligands. Science 2024, 384 (6692), 189–193.

116. Morteza Najarian, A.; Vafaie, M.; **Chen, B.**; García de Arquer, F. P.; Sargent, E. H. Photophysical Properties of Materials for High-Speed Photodetection. Nature Reviews Physics 2024, 1–12.

115. Xu, J.; Maxwell, A.; Song, Z.; Bati, A. S. R.; Chen, H.; Li, C.; Park, S. M.; Yan, Y.; **Chen, B.***; Sargent, E. H.* The Dynamic Adsorption Affinity of Ligands Is a Surrogate for the Passivation of Surface Defects. Nat. Commun. 2024, 15 (1), 2035.

114. Maxwell, A.; Chen, H.; Grater, L.; Li, C.; Teale, S.; Wang, J.; Zeng, L.; Wang, Z.; Park, S. M.; Vafaie, M.; Sidhik, S.; Metcalf, I. W.; Liu, Y.; Mohite, A. D.; **Chen, B.***; Sargent, E. H.* All-Perovskite Tandems Enabled by Surface Anchoring of Long-Chain Amphiphilic Ligands. ACS Energy Lett. 2024, 9, 520–527.

113. Yang, Y.; Liu, C.; Ding, Y.; Ding, B.; Xu, J.; Liu, A.; Yu, J.; Grater, L.; Zhu, H.; Hadke, S.; Sangwan, V.; Bati, A. S. R.; Hu, X.; Li, J.; Park, S. M.; Hersam, M.; **Chen, B.***; Nazeeruddin, M.*; Kanatzidis, M. G.*; Sargent, E. H.*A Thermotropic Liquid Crystal Enables Efficient and Stable Perovskite Solar Modules. Nat. Energy 2024, 9, 316–323.

112. Xu, F.; Aydin, E.; Liu, J.; Ugur, E.; Harrison, G. T.; Xu, L.; Vishal, B.; Yildirim, B. K.; Wang, M.; Ali, R.; Subbiah, A. S.; Yazmaciyan, A.; Zhumagali, S.; Yan, W.; Gao, Y.; Song, Z.; Li, C.; Fu, S.; **Chen, B.**; ur Rehman, A.; Babics, M.; Razzaq, A.; De Bastiani, M.; Allen, T. G.; Schwingenschlögl, U.; Yan, Y.; Laquai, F.; Sargent, E. H.; De Wolf, S. Monolithic Perovskite/Perovskite/Silicon Triple-Junction Solar Cells with Cation Double Displacement Enabled 2.0 eV Perovskites. Joule 2024, 8 (1), 224–240.

111. Liu, Y.†; Zhu, T.†; Grater, L.†; Chen, H.†; Reis, R.; Maxwell, A.; Cheng, M.; Dong, Y.; Teale, S.; Leontowich, A. F. G.; Kim, C.; Chan, P. T.; Wang, M.; Paritmongkol, W.; Gao, Y.; Park, S.; Xu, J.; Khan, J. I.; Laquai, F.; Walker, G. C.; Dravid, V. P.; **Chen, B.***; Sargent, E. H.* A Three-Dimensional Quantum Dot Network Stabilizes Perovskite Solids via Hydrostatic Strain. Matter 2024, 7 (1), 107–122.

110. Wang, J.; Zeng, L.; Zhang, D.; Maxwell, A.; Chen, H.; Datta, K.; Caiazzo, A.; Remmerswaal, W. H. M.; Schipper, N. R. M.; Chen, Z.; Ho, K.; Dasgupta, A.; Kusch, G.; Ollearo, R.; Bellini, L.; Hu, S.; Wang, Z.; Li, C.; Teale, S.; Grater, L.; **Chen, B.**; Wienk, M. M.; Oliver, R. A.; Snaith, H. J.; Janssen, R. A. J.; Sargent, E. H. Halide Homogenization for Low Energy Loss in 2-eV-Bandgap Perovskites and Increased Efficiency in All-Perovskite Triple-Junction Solar Cells. Nat. Energy 2023, 1–11.

109. Xu, J.; Chen, H.; Grater, L.; Liu, C.; Yang, Y.; Teale, S.; Maxwell, A.; Mahesh, S.; Wan, H.; Chang, Y.; **Chen, B.**; Rehl, B.; Park, S. M.; Kanatzidis, M. G.; Sargent, E. H. Anion Optimization for Bifunctional Surface Passivation in Perovskite Solar Cells. Nat. Mater. 2023, 1–8.

108. Liu, C.†; Yang, Y.†; Chen, H.†; Xu, J.†; Liu, A.†; Bati, A. S. R.; Zhu, H.; Grater, L.; Hadke, S. S.; Huang, C.; Sangwan, V. K.; Cai, T.; Shin, D.; Chen, L. X.; Hersam, M. C.; Mirkin, C. A.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Bimolecularly-passivated interface enables efficient and stable inverted perovskite solar cells, Science 2023, 382 (6672), 810–815.

107. Park, S. M.; Wei, M.; Lempesis, N.; Yu, W.; Hossain, T.; Agosta, L.; Carnevali, V.; Atapattu, H. R.; Serles, P.; Eickemeyer, F. T.; Shin, H.; Vafaie, M.; Choi, D.; Darabi, K.; Jung, E. D.; Yang, Y.; Kim, D. B.; Zakeeruddin, S. M.; **Chen, B.**; Amassian, A.; Filleter, T.; Kanatzidis, M. G.; Graham, K. R.; Xiao, L.; Rothlisberger, U.; Grätzel, M.; Sargent, E. H. Low-Loss Contacts on Textured Substrates for Inverted Perovskite Solar Cells. Nature 2023, 1–3.

106. Zhu, H.; Teale, S.; Lintangpradipto, M. N.; Mahesh, S.; **Chen, B.**; McGehee, M. D.; Sargent, E. H.; Bakr, O. M. Long-Term Operating Stability in Perovskite Photovoltaics. Nat. Rev. Mater. 2023, 1–18.

105. Park, S. M.; Wei, M.; Xu, J.; Atapattu, H. R.; Eickemeyer, F. T.; Darabi, K.; Grater, L.; Yang, Y.; Liu, C.; Teale, S.; **Chen, B.**; Chen, H.; Wang, T.; Zeng, L.; Maxwell, A.; Wang, Z.; Rao, K. R.; Cai, Z.; Zakeeruddin, S. M.; Pham, J. T.; Risko, C. M.; Amassian, A.; Kanatzidis, M.

G.; Graham, K. R.; Grätzel, M.; Sargent, E. H. Engineering Ligand Reactivity Enables High-Temperature Operation of Stable Perovskite Solar Cells. Science 2023, 381 (6654), 209–215.

104. Grater, L.; Wang, M.; Teale, S.; Mahesh, S.; Maxwell, A.; Liu, Y.; Park, S. M.; **Chen, B.**; Laquai, F.; Kanatzidis, M. G.; Sargent, E. H. Sterically Suppressed Phase Segregation in 3D Hollow Mixed-Halide Wide Band Gap Perovskites. J. Phys. Chem. Lett. 2023, 14 (26), 6157–6162.

103. Wang, Z.; Zeng, L.; Zhu, T.; Chen, H.; **Chen, B.**; Kubicki, D. J.; Balvanz, A.; Li, C.; Maxwell, A.; Ugur, E.; Dos Reis, R.; Cheng, M.; Yang, G.; Subedi, B.; Luo, D.; Hu, J.; Wang, J.; Teale, S.; Mahesh, S.; Wang, S.; Hu, S.; Jung, E. D.; Wei, M.; Park, S. M.; Grater, L.; Aydin, E.; Song, Z.; Podraza, N. J.; Lu, Z.-H.; Huang, J.; Dravid, V. P.; De Wolf, S.; Yan, Y.; Grätzel, M.; Kanatzidis, M. G.; Sargent, E. H. Suppressed Phase Segregation for Triple-Junction Perovskite Solar Cells. Nature 2023, 618 (7963), 74–79.

102. Ugur, E.; Aydin, E.; De Bastiani, M.; Harrison, G. T.; Yildirim, B. K.; Teale, S.; **Chen, B.**; Liu, J.; Wang, M.; Seitkhan, A.; Babics, M.; Subbiah, A. S.; Said, A. A.; Azmi, R.; Rehman, A. ur; Allen, T. G.; Schulz, P.; Sargent, E. H.; Laquai, F.; De Wolf, S. Front-Contact Passivation through 2D/3D Perovskite Heterojunctions Enables Efficient Bifacial Perovskite/Silicon Tandem Solar Cells. Matter 2023.

101. Li, T.; Xu, J.; Lin, R.; Teale, S.; Li, H.; Liu, Z.; Duan, C.; Zhao, Q.; Xiao, K.; Wu, P.; **Chen, B.**; Jiang, S.; Xiong, S.; Luo, H.; Wan, S.; Li, L.; Bao, Q.; Tian, Y.; Gao, X.; Xie, J.; Sargent, E. H.; Tan, H. Inorganic Wide-Bandgap Perovskite Subcells with Dipole Bridge for All-Perovskite Tandems. Nat. Energy 2023, 8 (6), 610–620.

100. Lee, M. G.; Li, X.-Y.; Ozden, A.; Wicks, J.; Ou, P.; Li, Y.; Dorakhan, R.; Lee, J.; Park, H. K.; Yang, J. W.; **Chen, B.**; Abed, J.; dos Reis, R.; Lee, G.; Huang, J. E.; Peng, T.; Chin, Y.-H.; Sinton, D.; Sargent, E. H. Selective Synthesis of Butane from Carbon Monoxide Using Cascade Electrolysis and Thermocatalysis at Ambient Conditions. Nat. Catal. 2023, 6 (4), 310–318.

99. Luo, M.; Ozden, A.; Wang, Z.; Li, F.; Erick Huang, J.; Hung, S.-F.; Wang, Y.; Li, J.; Nam, D.-H.; Li, Y. C.; Xu, Y.; Lu, R.; Zhang, S.; Lum, Y.; Ren, Y.; Fan, L.; Wang, F.; Li, H.-H.; Appadoo, D.; Dinh, C.-T.; Liu, Y.; **Chen, B.**; Wicks, J.; Chen, H.; Sinton, D.; Sargent, E. H. Coordination Polymer Electrocatalysts Enable Efficient CO-to-Acetate Conversion. Adv. Mater. 2023, 35 (10), e2209567.

98. Li, C.; Wang, X.; Bi, E.; Jiang, F.; Park, S. M.; Li, Y.; Chen, L.; Wang, Z.; Zeng, L.; Chen, H.; Liu, Y.; Grice, C. R.; Abudulimu, A.; Chung, J.; Xian, Y.; Zhu, T.; Lai, H.; **Chen, B.**; Ellingson, R. J.; Fu, F.; Ginger, D. S.; Song, Z.; Sargent, E. H.; Yan, Y. Rational Design of Lewis Base Molecules for Stable and Efficient Inverted Perovskite Solar Cells. Science 2023, 379 (6633), 690–694.

97. Luo, L.; Zeng, H.; Wang, Z.; Li, M.; You, S.; **Chen, B.**; Maxwell, A.; An, Q.; Cui, L.; Luo, D.; Hu, J.; Li, S.; Cai, X.; Li, W.; Li, L.; Guo, R.; Huang, R.; Liang, W.; Lu, Z.-H.; Mai, L.; Rong, Y.; Sargent, E. H.; Li, X. Stabilization of 3D/2D Perovskite Heterostructures via Inhibition of Ion Diffusion by Cross-Linked Polymers for Solar Cells with Improved Performance. Nat. Energy 2023, 8 (3), 294–303.

96. Zhu, T.; Teale, S.; Grater, L.; Vasileiadou, E. S.; Sharir-Smith, J.; **Chen, B.**; Kanatzidis, M. G.; Sargent, E. H. Coupling Photogeneration with Thermodynamic Modeling of Light-Induced Alloy Segregation Enables the Discovery of Stabilizing Dopants. arXiv [cond-mat.mtrl-sci], 2023.

95. Chen, H.; Maxwell, A.; Li, C.; Teale, S.; **Chen, B.**; Zhu, T.; Ugur, E.; Harrison, G.; Grater, L.; Wang, J.; Wang, Z.; Zeng, L.; Park, S. M.; Chen, L.; Serles, P.; Awni, R. A.; Subedi, B.; Zheng, X.; Xiao, C.; Podraza, N. J.; Filleter, T.; Liu, C.; Yang, Y.; Luther, J. M.; De Wolf, S.; Kanatzidis, M. G.; Yan, Y.; Sargent, E. H. Regulating Surface Potential Maximizes Voltage in All-Perovskite Tandems. Nature 2023, 613 (7945), 676–681.

94. Mahesh, S.; **Chen, B.**; Sargent, E. H. All-Perovskite Tandems Go Bifacial. Light Sci. Appl. 2023, 12 (1), 13.

93. Steele, J. A.; Braeckevelt, T.; Prakasam, V.; Degutis, G.; Yuan, H.; Jin, H.; Solano, E.; Puech, P.; Basak, S.; Pintor-Monroy, M. I.; Van Gorp, H.; Fleury, G.; Yang, R. X.; Lin, Z.; Huang, H.; Debroye, E.; Chernyshov, D.; **Chen, B.**; Wei, M.; Hou, Y.; Gehlhaar, R.; Genoe, J.; De Feyter, S.; Rogge, S. M. J.; Walsh, A.; Sargent, E. H.; Yang, P.; Hofkens, J.; Van Speybroeck, V.; Roeffaers, M. B. J. An Embedded Interfacial Network Stabilizes Inorganic CsPbI3 Perovskite Thin Films. Nat. Commun. 2022, 13 (1), 7513.

92. Lee, S.; Park, S. M.; Jung, E. D.; Zhu, T.; Pina, J. M.; Anwar, H.; Wu, F.-Y.; Chen, G.-L.; Dong, Y.; Cui, T.; Wei, M.; Bertens, K.; Wang, Y.-K.; **Chen, B.**; Filleter, T.; Hung, S.-F.; Won, Y.-H.; Kim, K. H.; Hoogland, S.; Sargent, E. H. Dipole Engineering through the Orientation of Interface Molecules for Efficient InP Quantum Dot Light-Emitting Diodes. J. Am. Chem. Soc. 2022, 144 (45), 20923–20930.

91. Zhou, Y.; Zhou, S.; Ying, P.; Zhao, Q.; Xie, Y.; Gong, M.; Jiang, P.; Cai, H.; **Chen, B.**; Tongay, S.; Zhang, J.; Jie, W.; Wang, T.; Tan, P.; Liu, D.; Kuball, M. Unusual Deformation and Fracture in Gallium Telluride Multilayers. J. Phys. Chem. Lett. 2022, 13 (17), 3831–3839.

90. **Chen, B.**; Sargent, E. H. What Does Net Zero by 2050 Mean to the Solar Energy Materials Researcher? Matter 2022, 5 (5), 1322–1325.

89. Chen, H.; Teale, S.; **Chen, B.**; Hou, Y.; Grater, L.; Zhu, T.; Bertens, K.; Park, S. M.; Atapattu, H. R.; Gao, Y.; Wei, M.; Johnston, A. K.; Zhou, Q.; Xu, K.; Yu, D.; Han, C.; Cui, T.; Jung, E. H.; Zhou, C.; Zhou, W.; Proppe, A. H.; Hoogland, S.; Laquai, F.; Filleter, T.; Graham, K. R.; Ning, Z.; Sargent, E. H. Quantum-Size-Tuned Heterostructures Enable Efficient and Stable Inverted Perovskite Solar Cells. Nat. Photonics 2022, 16 (5), 352–358.

88. Liu, Y.; Li, Z.; Xu, J.; Dong, Y.; **Chen, B.**; Park, S. M.; Ma, D.; Lee, S.; Huang, J. E.; Teale, S.; Voznyy, O.; Sargent, E. H. Wide-Bandgap Perovskite Quantum Dots in Perovskite Matrix for Sky-Blue Light-Emitting Diodes. J. Am. Chem. Soc. 2022, 144 (9), 4009–4016.

87. Lin, R.; Xu, J.; Wei, M.; Wang, Y.; Qin, Z.; Liu, Z.; Wu, J.; Xiao, K.; **Chen, B.**; Park, S. M.; Chen, G.; Atapattu, H. R.; Graham, K. R.; Xu, J.; Zhu, J.; Li, L.; Zhang, C.; Sargent, E. H.; Tan, H. All-Perovskite Tandem Solar Cells with Improved Grain Surface Passivation. Nature 2022, 603 (7899), 73–78.

86. Xu, J.; Maxwell, A.; Wei, M.; Wang, Z.; **Chen, B.**; Zhu, T.; Sargent, E. H. Defect Tolerance of Mixed B-Site Organic–Inorganic Halide Perovskites. ACS Energy Lett. 2021, 6 (12), 4220–4227.

85. Fang, Z.; Wang, L.; Mu, X.; **Chen, B.**; Xiong, Q.; Wang, W. D.; Ding, J.; Gao, P.; Wu, Y.; Cao, J. Grain Boundary Engineering with Self-Assembled Porphyrin Supramolecules for Highly Efficient Large-Area Perovskite Photovoltaics. J. Am. Chem. Soc. 2021, 143 (45), 18989–18996.

84. Ma, D.; Lin, K.; Dong, Y.; Choubisa, H.; Proppe, A. H.; Wu, D.; Wang, Y.-K.; **Chen, B.**; Li, P.; Fan, J. Z.; Yuan, F.; Johnston, A.; Liu, Y.; Kang, Y.; Lu, Z.-H.; Wei, Z.; Sargent, E. H. Distribution Control Enables Efficient Reduced-Dimensional Perovskite LEDs. Nature 2021, 599 (7886), 594–598.

83. Peng, T.; Zhuang, T.; Yan, Y.; Qian, J.; Dick, G. R.; Behaghel de Bueren, J.; Hung, S.-F.; Zhang, Y.; Wang, Z.; Wicks, J.; Garcia de Arquer, F. P.; Abed, J.; Wang, N.; Sedighian Rasouli, A.; Lee, G.; Wang, M.; He, D.; Wang, Z.; Liang, Z.; Song, L.; Wang, X.; **Chen, B.**; Ozden, A.; Lum, Y.; Leow, W. R.; Luo, M.; Meira, D. M.; Ip, A. H.; Luterbacher, J. S.; Zhao, W.; Sargent, E. H. Ternary Alloys Enable Efficient Production of Methoxylated Chemicals via Selective Electrocatalytic Hydrogenation of Lignin Monomers. J. Am. Chem. Soc. 2021, 143 (41), 17226–17235.

82. **Chen, B.**; Chen, H.; Hou, Y.; Xu, J.; Teale, S.; Bertens, K.; Chen, H.; Proppe, A.; Zhou, Q.; Yu, D.; Xu, K.; Vafaie, M.; Liu, Y.; Dong, Y.; Jung, E. H.; Zheng, C.; Zhu, T.; Ning, Z.; Sargent, E. H. Passivation of the Buried Interface via Preferential Crystallization of 2D Perovskite on Metal Oxide Transport Layers. Adv. Mater. 2021, 33 (41), e2103394.

81. Zhou, C.; M Pina, J.; Zhu, T.; H Parmar, D.; Chang, H.; Yu, J.; Yuan, F.; Bappi, G.; Hou, Y.; Zheng, X.; Abed, J.; Chen, H.; Zhang, J.; Gao, Y.; **Chen, B.**; Wang, Y.-K.; Chen, H.; Zhang, T.; Hoogland, S.; Saidaminov, M. I.; Sun, L.; Bakr, O. M.; Dong, H.; Zhang, L.; H Sargent, E. Quantum Dot Self-Assembly Enables Low-Threshold Lasing. Adv. Sci. (Weinh.) 2021, 8 (20), e2101125.

80. Liu, Y.; Dong, Y.; Zhu, T.; Ma, D.; Proppe, A.; **Chen, B.**; Zheng, C.; Hou, Y.; Lee, S.; Sun, B.; Jung, E. H.; Yuan, F.; Wang, Y.-K.; Sagar, L. K.; Hoogland, S.; García de Arquer, F. P.; Choi, M.-J.; Singh, K.; Kelley, S. O.; Voznyy, O.; Lu, Z.-H.; Sargent, E. H. Bright and Stable Light-Emitting Diodes Based on Perovskite Quantum Dots in Perovskite Matrix. J. Am. Chem. Soc. 2021, 143 (38), 15606–15615.

79. Li, H.; Wines, D.; **Chen, B.**; Yumigeta, K.; Sayyad, Y. M.; Kopaszek, J.; Yang, S.; Ataca, C.; Sargent, E. H.; Tongay, S. Abnormal Phase Transition and Band Renormalization of Guanidinium-Based Organic-Inorganic Hybrid Perovskite. ACS Appl. Mater. Interfaces 2021, 13 (37), 44964–44971.

78. De Bastiani, M.; Van Kerschaver, E.; Jeangros, Q.; Ur Rehman, A.; Aydin, E.; Isikgor, F. H.; Mirabelli, A. J.; Babics, M.; Liu, J.; Zhumagali, S.; Ugur, E.; Harrison, G. T.; Allen, T. G.; **Chen, B.**; Hou, Y.; Shikin, S.; Sargent, E. H.; Ballif, C.; Salvador, M.; De Wolf, S. Toward Stable Monolithic Perovskite/Silicon Tandem Photovoltaics: A Six-Month Outdoor Performance Study in a Hot and Humid Climate. ACS Energy Lett. 2021, 6 (8), 2944–2951.

77. Biondi, M.; Choi, M.-J.; Wang, Z.; Wei, M.; Lee, S.; Choubisa, H.; Sagar, L. K.; Sun, B.; Baek, S.-W.; **Chen, B.**; Todorović, P.; Najarian, A. M.; Sedighian Rasouli, A.; Nam, D.-H.; Vafaie, M.; Li, Y. C.; Bertens, K.; Hoogland, S.; Voznyy, O.; García de Arquer, F. P.; Sargent, E. H. Facet-Oriented Coupling Enables Fast and Sensitive Colloidal Quantum Dot Photodetectors. Adv. Mater. 2021, 33 (33), e2101056.

76. Aydin, E.; Liu, J.; Ugur, E.; Azmi, R.; Harrison, G. T.; Hou, Y.; **Chen, B.**; Zhumagali, S.; De Bastiani, M.; Wang, M.; Raja, W.; Allen, T. G.; Rehman, A. ur; Subbiah, A. S.; Babics, M.; Babayigit, A.; Isikgor, F. H.; Wang, K.; Van Kerschaver, E.; Tsetseris, L.; Sargent, E. H.; Laquai, F.; De Wolf, S. Ligand-Bridged Charge Extraction and Enhanced Quantum Efficiency Enable Efficient n–i–p Perovskite/Silicon Tandem Solar Cells. Energy Environ. Sci. 2021, 14 (8), 4377–4390.

75. Wang, Y.-K.; Yuan, F.; Dong, Y.; Li, J.-Y.; Johnston, A.; **Chen, B.**; Saidaminov, M. I.; Zhou, C.; Zheng, X.; Hou, Y.; Bertens, K.; Ebe, H.; Ma, D.; Deng, Z.; Yuan, S.; Chen, R.; Sagar, L. K.; Liu, J.; Fan, J.; Li, P.; Li, X.; Gao, Y.; Fung, M.-K.; Lu, Z.-H.; Bakr, O. M.; Liao, L.-S.; Sargent, E. H. All-Inorganic Quantum-Dot LEDs Based on a Phase-Stabilized -CsPbI3 Perovskite. Angew. Chem. Int. Ed Engl. 2021, 60 (29), 16164–16170.

74. Li, J.; Ozden, A.; Wan, M.; Hu, Y.; Li, F.; Wang, Y.; Zamani, R. R.; Ren, D.; Wang, Z.; Xu, Y.; Nam, D.-H.; Wicks, J.; **Chen, B.**; Wang, X.; Luo, M.; Graetzel, M.; Che, F.; Sargent, E. H.; Sinton, D. Silica-Copper Catalyst Interfaces Enable Carbon-Carbon Coupling towards Ethylene Electrosynthesis. Nat. Commun. 2021, 12 (1), 2808.

73. Aydin, E.; De Wolf, S.; Subbiah, A. S.; Liu, J.; Ugur, E.; Azmi, R.; Allen, T.; de Bastiani, M.; Babics, M.; Isikgor, F. H.; **Chen, B.**; Hou, Y.; Laquai, F.; Sargent, E. H.; Rehman, A. U. The Multiple Ways of Making Perovskite/Silicon Tandem Solar Cells: Which Way to Go? 2021.

72. Huang, Z.; Wei, M.; Proppe, A. H.; Chen, H.; **Chen, B.**; Hou, Y.; Ning, Z.; Sargent, E. Band

Engineering via Gradient Molecular Dopants for CsFA Perovskite Solar Cells. Adv. Funct. Mater. 2021, 31 (18), 2010572.

71. De Bastiani, M.; Mirabelli, A. J.; Hou, Y.; Gota, F.; Aydin, E.; Allen, T. G.; Troughton, J.; Subbiah, A. S.; Isikgor, F. H.; Liu, J.; Xu, L.; **Chen, B.**; Van Kerschaver, E.; Baran, D.; Fraboni, B.; Salvador, M. F.; Paetzold, U. W.; Sargent, E. H.; De Wolf, S. Efficient Bifacial Monolithic Perovskite/Silicon Tandem Solar Cells via Bandgap Engineering. Nat. Energy 2021, 6 (2), 167–175.

70. Brasington, A.; Golla, D.; Dave, A.; **Chen, B.**; Tongay, S.; Schaibley, J.; LeRoy, B. J.; Sandhu, A. Role of Defects and Phonons in Bandgap Dynamics of Monolayer WS2 at High Carrier Densities. JPhys Mater. 2021, 4 (1), 015005.

69. Nam, D.-H.; Shekhah, O.; Lee, G.; Mallick, A.; Jiang, H.; Li, F.; **Chen, B.**; Wicks, J.; Eddaoudi, M.; Sargent, E. H. Intermediate Binding Control Using Metal-Organic Frameworks Enhances Electrochemical CO2 Reduction. J. Am. Chem. Soc. 2020, 142 (51), 21513–21521.

68. Li, Y.; Xu, A.; Lum, Y.; Wang, X.; Hung, S.-F.; **Chen, B.**; Wang, Z.; Xu, Y.; Li, F.; Abed, J.; Huang, J. E.; Rasouli, A. S.; Wicks, J.; Sagar, L. K.; Peng, T.; Ip, A. H.; Sinton, D.; Jiang, H.; Li, C.; Sargent, E. H. Promoting CO2 Methanation via Ligand-Stabilized Metal Oxide Clusters as Hydrogen-Donating Motifs. Nat. Commun. 2020, 11 (1), 6190.

67. Lee, S.; Sagar, L. K.; Li, X.; Dong, Y.; **Chen, B.**; Gao, Y.; Ma, D.; Levina, L.; Grenville, A.; Hoogland, S.; García de Arquer, F. P.; Sargent, E. H. InP-Quantum-Dot-in-ZnS-Matrix Solids for Thermal and Air Stability. Chem. Mater. 2020, 32 (22), 9584–9590.

66. Yuan, F.; Zheng, X.; Johnston, A.; Wang, Y.-K.; Zhou, C.; Dong, Y.; **Chen, B.**; Chen, H.; Fan, J. Z.; Sharma, G.; Li, P.; Gao, Y.; Voznyy, O.; Kung, H.-T.; Lu, Z.-H.; Bakr, O. M.; Sargent, E. H. Color-Pure Red Light-Emitting Diodes Based on Two-Dimensional Lead-Free Perovskites. Sci. Adv. 2020, 6 (42), eabb0253.

65. Kim, H. I.; Baek, S.-W.; Choi, M.-J.; **Chen, B.**; Ouellette, O.; Choi, K.; Scheffel, B.; Choi, H.; Biondi, M.; Hoogland, S.; García de Arquer, F. P.; Park, T.; Sargent, E. H. Monolithic Organic/Colloidal Quantum Dot Hybrid Tandem Solar Cells via Buffer Engineering. Adv. Mater. 2020, 32 (42), e2004657.

64. Lee, S.; Choi, M.-J.; Sharma, G.; Biondi, M.; **Chen, B.**; Baek, S.-W.; Najarian, A. M.; Vafaie, M.; Wicks, J.; Sagar, L. K.; Hoogland, S.; de Arquer, F. P. G.; Voznyy, O.; Sargent, E. H. Orthogonal Colloidal Quantum Dot Inks Enable Efficient Multilayer Optoelectronic Devices. Nat. Commun. 2020, 11 (1), 4814.

63. Sagar, L. K.; Bappi, G.; Johnston, A.; **Chen, B.**; Todorović, P.; Levina, L.; Saidaminov, M. I.; García de Arquer, F. P.; Nam, D.-H.; Choi, M.-J.; Hoogland, S.; Voznyy, O.; Sargent, E. H. Suppression of Auger Recombination by Gradient Alloying in InAs/CdSe/CdS QDs. Chem. Mater. 2020, 32 (18), 7703–7709.

62. Ozden, A.; Li, F.; García de Arquer, F. P.; Rosas-Hernández, A.; Thevenon, A.; Wang, Y.; Hung, S.-F.; Wang, X.; **Chen, B.**; Li, J.; Wicks, J.; Luo, M.; Wang, Z.; Agapie, T.; Peters, J. C.; Sargent, E. H.; Sinton, D. High-Rate and Efficient Ethylene Electrosynthesis Using a Catalyst/Promoter/Transport Layer. ACS Energy Lett. 2020, 5 (9), 2811–2818.

61. Jung, E. H.; **Chen, B.**; Bertens, K.; Vafaie, M.; Teale, S.; Proppe, A.; Hou, Y.; Zhu, T.; Zheng, C.; Sargent, E. H. Bifunctional Surface Engineering on SnO2 Reduces Energy Loss in Perovskite Solar Cells. ACS Energy Lett. 2020, 5 (9), 2796–2801.

60. Dong, Y.; Wang, Y.-K.; Yuan, F.; Johnston, A.; Liu, Y.; Ma, D.; Choi, M.-J.; **Chen, B.**; Chekini, M.; Baek, S.-W.; Sagar, L. K.; Fan, J.; Hou, Y.; Wu, M.; Lee, S.; Sun, B.; Hoogland, S.; Quintero-Bermudez, R.; Ebe, H.; Todorovic, P.; Dinic, F.; Li, P.; Kung, H. T.; Saidaminov,

M. I.; Kumacheva, E.; Spiecker, E.; Liao, L.-S.; Voznyy, O.; Lu, Z.-H.; Sargent, E. H. Bipolar-Shell Resurfacing for Blue LEDs Based on Strongly Confined Perovskite Quantum Dots. Nat. Nanotechnol. 2020, 15 (8), 668–674.

59. Li, J.; Xu, A.; Li, F.; Wang, Z.; Zou, C.; Gabardo, C. M.; Wang, Y.; Ozden, A.; Xu, Y.; Nam, D.-H.; Lum, Y.; Wicks, J.; **Chen, B.**; Wang, Z.; Chen, J.; Wen, Y.; Zhuang, T.; Luo, M.; Du, X.; Sham, T.-K.; Zhang, B.; Sargent, E. H.; Sinton, D. Enhanced Multi-Carbon Alcohol Electroproduction from CO via Modulated Hydrogen Adsorption. Nature communications 2020, 11 (1), 3685.

58. Wang, Y.-K.; Ma, D.; Yuan, F.; Singh, K.; Pina, J. M.; Johnston, A.; Dong, Y.; Zhou, C.; **Chen, B.**; Sun, B.; Ebe, H.; Fan, J.; Sun, M.-J.; Gao, Y.; Lu, Z.-H.; Voznyy, O.; Liao, L.-S.; Sargent, E. H. Chelating-Agent-Assisted Control of CsPbBr3 Quantum Well Growth Enables Stable Blue Perovskite Emitters. Nat. Commun. 2020, 11 (1), 3674.

57. Fan, J. Z.; Vafaie, M.; Bertens, K.; Sytnyk, M.; Pina, J. M.; Sagar, L. K.; Ouellette, O.; Proppe, A. H.; Rasouli, A. S.; Gao, Y.; Baek, S.-W.; **Chen, B.**; Laquai, F.; Hoogland, S.; Arquer, F. P. G. de; Heiss, W.; Sargent, E. H. Micron Thick Colloidal Quantum Dot Solids. Nano Lett. 2020, 20 (7), 5284–5291.

56. Huang, Z.; **Chen, B.**; Sagar, L. K.; Hou, Y.; Proppe, A.; Kung, H.-T.; Yuan, F.; Johnston, A.; Saidaminov, M. I.; Jung, E. H.; Lu, Z.-H.; Kelley, S. O.; Sargent, E. H. Stable, Bromine-Free, Tetragonal Perovskites with 1.7 eV Bandgaps via A-Site Cation Substitution. ACS Mater. Lett. 2020, 2 (7), 869–872.

55. Teale, S.; Proppe, A. H.; Jung, E. H.; Johnston, A.; Parmar, D. H.; **Chen, B.**; Hou, Y.; Kelley, S. O.; Sargent, E. H. Dimensional Mixing Increases the Efficiency of 2D/3D Perovskite Solar Cells. J. Phys. Chem. Lett. 2020, 11 (13), 5115–5119.

54. Leow, W. R.; Lum, Y.; Ozden, A.; Wang, Y.; Nam, D.-H.; **Chen, B.**; Wicks, J.; Zhuang, T.-T.; Li, F.; Sinton, D.; Sargent, E. H. Chloride-Mediated Selective Electrosynthesis of Ethylene and Propylene Oxides at High Current Density. Science 2020, 368 (6496), 1228–1233.

53. Chen, H.; Pina, J. M.; Yuan, F.; Johnston, A.; Ma, D.; **Chen, B.**; Li, Z.; Dumont, A.; Li, X.; Liu, Y.; Hoogland, S.; Zajacz, Z.; Lu, Z.; Sargent, E. H. Multiple Self-Trapped Emissions in the Lead-Free Halide Cs3Cu2I5. J. Phys. Chem. Lett. 2020, 11 (11), 4326–4330.

52. Wang, X.; Wang, Z.; de Arquer, F. P. G.; Dinh, C.-T.; Ozden, A.; Li, Y. C.; Nam, D.-H.; Li, J.; Liu, Y.-S.; Wicks, J.; Chen, Z.; Chi, M.; **Chen, B.**; Wang, Y.; Tam, J.; Howe, J. Y.; Proppe, A.; Todorović, P.; Li, F.; Zhuang, T.-T.; Gabardo, C. M.; Kirmani, A. R.; McCallum, C.; Hung, S.-F.; Lum, Y.; Luo, M.; Min, Y.; Xu, A.; O'Brien, C. P.; Stephen, B.; Sun, B.; Ip, A. H.; Richter, L. J.; Kelley, S. O.; Sinton, D.; Sargent, E. H. Efficient Electrically Powered CO2-to-Ethanol via Suppression of Deoxygenation. Nature Energy 2020, 5 (6), 478–486.

51. Sun, B.; Vafaie, M.; Levina, L.; Wei, M.; Dong, Y.; Gao, Y.; Kung, H. T.; Biondi, M.; Proppe, A. H.; **Chen, B.**; Choi, M.-J.; Sagar, L. K.; Voznyy, O.; Kelley, S. O.; Laquai, F.; Lu, Z.-H.; Hoogland, S.; García de Arquer, F. P.; Sargent, E. H. Ligand-Assisted Reconstruction of Colloidal Quantum Dots Decreases Trap State Density. Nano Lett. 2020, 20 (5), 3694–3702.

50. Sagar, L. K.; Bappi, G.; Johnston, A.; **Chen, B.**; Todorović, P.; Levina, L.; Saidaminov, M. I.; García de Arquer, F. P.; Hoogland, S.; Sargent, E. H. Single-Precursor Intermediate Shelling Enables Bright, Narrow Line Width InAs/InZnP-Based QD Emitters. Chem. Mater. 2020, 32 (7), 2919–2925.

49. Wu, K.; Blei, M.; **Chen, B.**; Liu, L.; Cai, H.; Brayfield, C.; Wright, D.; Zhuang, H.; Tongay, S. Phase Transition across Anisotropic NbS3 and Direct Gap Semiconductor TiS3 at Nominal Titanium Alloying Limit. Adv. Mater. 2020, 32 (17), e2000018.

48. Liang, H.; Yuan, F.; Johnston, A.; Gao, C.; Choubisa, H.; Gao, Y.; Wang, Y.-K.; Sagar, L. K.; Sun, B.; Li, P.; Bappi, G.; **Chen, B.**; Li, J.; Wang, Y.; Dong, Y.; Ma, D.; Gao, Y.; Liu, Y.; Yuan, M.; Saidaminov, M. I.; Hoogland, S.; Lu, Z.-H.; Sargent, E. H. High Color Purity Lead-Free Perovskite Light-Emitting Diodes via Sn Stabilization. Adv. Sci. (Weinh.) 2020, 7 (8), 1903213.

47. Xue, D.-J.; Hou, Y.; Liu, S.-C.; Wei, M.; **Chen, B.**; Huang, Z.; Li, Z.; Sun, B.; Proppe, A. H.; Dong, Y.; Saidaminov, M. I.; Kelley, S. O.; Hu, J.-S.; Sargent, E. H. Regulating Strain in Perovskite Thin Films through Charge-Transport Layers. Nat. Commun. 2020, 11 (1), 1514.

46. Ma, D.; Todorović, P.; Meshkat, S.; Saidaminov, M. I.; Wang, Y.-K.; **Chen, B.**; Li, P.; Scheffel, B.; Quintero-Bermudez, R.; Fan, J. Z.; Dong, Y.; Sun, B.; Xu, C.; Zhou, C.; Hou, Y.; Li, X.; Kang, Y.; Voznyy, O.; Lu, Z.-H.; Ban, D.; Sargent, E. H. Chloride Insertion-Immobilization Enables Bright, Narrowband, and Stable Blue-Emitting Perovskite Diodes. J. Am. Chem. Soc. 2020, 142 (11), 5126–5134.

45. **Chen, B.**; Baek, S.-W.; Hou, Y.; Aydin, E.; De Bastiani, M.; Scheffel, B.; Proppe, A.; Huang, Z.; Wei, M.; Wang, Y.-K.; Jung, E.-H.; Allen, T. G.; Van Kerschaver, E.; García de Arquer, F. P.; Saidaminov, M. I.; Hoogland, S.; De Wolf, S.; Sargent, E. H. Enhanced Optical Path and Electron Diffusion Length Enable High-Efficiency Perovskite Tandems. Nat. Commun. 2020, 11 (1), 1257.

44. Hou, Y.; Aydin, E.; De Bastiani, M.; Xiao, C.; Isikgor, F. H.; Xue, D.-J.; **Chen, B.**; Chen, H.; Bahrami, B.; Chowdhury, A. H.; Johnston, A.; Baek, S.-W.; Huang, Z.; Wei, M.; Dong, Y.; Troughton, J.; Jalmood, R.; Mirabelli, A. J.; Allen, T. G.; Van Kerschaver, E.; Saidaminov, M. I.; Baran, D.; Qiao, Q.; Zhu, K.; De Wolf, S.; Sargent, E. H. Efficient Tandem Solar Cells with Solution-Processed Perovskite on Textured Crystalline Silicon. Science 2020, 367 (6482), 1135–1140.

43. Yuan, F.; Wang, Y.-K.; Sharma, G.; Dong, Y.; Zheng, X.; Li, P.; Johnston, A.; Bappi, G.; Fan, J. Z.; Kung, H.; **Chen, B.**; Saidaminov, M. I.; Singh, K.; Voznyy, O.; Bakr, O. M.; Lu, Z.-H.; Sargent, E. H. Bright High-Colour-Purity Deep-Blue Carbon Dot Light-Emitting Diodes via Efficient Edge Amination. Nat. Photonics 2020, 14 (3), 171–176.

42. Zheng, X.; Hou, Y.; Bao, C.; Yin, J.; Yuan, F.; Huang, Z.; Song, K.; Liu, J.; Troughton, J.; Gasparini, N.; Zhou, C.; Lin, Y.; Xue, D.-J.; **Chen, B.**; Johnston, A. K.; Wei, N.; Hedhili, M. N.; Wei, M.; Alsalloum, A. Y.; Maity, P.; Turedi, B.; Yang, C.; Baran, D.; Anthopoulos, T. D.; Han, Y.; Lu, Z.-H.; Mohammed, O. F.; Gao, F.; Sargent, E. H.; Bakr, O. M. Managing Grains and Interfaces via Ligand Anchoring Enables 22.3

41. Lum, Y.; Huang, J. E.; Wang, Z.; Luo, M.; Nam, D.-H.; Leow, W. R.; **Chen, B.**; Wicks, J.; Li, Y. C.; Wang, Y.; Dinh, C.-T.; Li, J.; Zhuang, T.-T.; Li, F.; Sham, T.-K.; Sinton, D.; Sargent, E. H. Tuning OH Binding Energy Enables Selective Electrochemical Oxidation of Ethylene to Ethylene Glycol. Nat. Catal. 2020, 3 (1), 14–22.

40. Johnston, A.; Dinic, F.; Todorović, P.; **Chen, B.**; Sagar, L. K.; Saidaminov, M. I.; Hoogland, S.; Voznyy, O.; Sargent, E. H. Narrow Emission from Rb 3 Sb 2 I 9 Nanoparticles. Adv. Opt. Mater. 2020, 8 (1), 1901606.

39. Luo, M.; Wang, Z.; Li, Y. C.; Li, J.; Li, F.; Lum, Y.; Nam, D.-H.; **Chen, B.**; Wicks, J.; Xu, A.; Zhuang, T.; Leow, W. R.; Wang, X.; Dinh, C.-T.; Wang, Y.; Wang, Y.; Sinton, D.; Sargent, E. H. Hydroxide Promotes Carbon Dioxide Electroreduction to Ethanol on Copper via Tuning of Adsorbed Hydrogen. Nat. Commun. 2019, 10 (1), 5814.

38. Li, F.; Li, Y. C.; Wang, Z.; Li, J.; Nam, D.-H.; Lum, Y.; Luo, M.; Wang, X.; Ozden, A.; Hung, S.-F.; **Chen, B.**; Wang, Y.; Wicks, J.; Xu, Y.; Li, Y.; Gabardo, C. M.; Dinh, C.-T.; Wang, Y.; Zhuang, T.-T.; Sinton, D.; Sargent, E. H. Cooperative CO2-to-Ethanol Conversion via Enriched Intermediates at Molecule–Metal Catalyst Interfaces. Nat. Catal. 2019, 3 (1), 75–82.

37. Todorović, P.; Ma, D.; **Chen, B.**; Quintero-Bermudez, R.; Saidaminov, M. I.; Dong, Y.; Lu, Z.-H.; Sargent, E. H. Spectrally Tunable and Stable Electroluminescence Enabled by Rubidium Doping of CsPbBr 3 Nanocrystals. Adv. Opt. Mater. 2019, 7 (24), 1901440.

36. Li, H.; Wu, K.; Yang, S.; Boland, T.; **Chen, B.**; Singh, A. K.; Tongay, S. Anomalous Phase Transition Behavior in Hydrothermal Grown Layered Tellurene. Nanoscale 2019, 11 (42), 20245–20251.

35. Mercado, E.; Zhou, Y.; Xie, Y.; Zhao, Q.; Cai, H.; **Chen, B.**; Jie, W.; Tongay, S.; Wang, T.; Kuball, M. Passivation of Layered Gallium Telluride by Double Encapsulation with Graphene. ACS Omega 2019, 4 (19), 18002–18010.

34. Zhuang, T.-T.; Nam, D.-H.; Wang, Z.; Li, H.-H.; Gabardo, C. M.; Li, Y.; Liang, Z.-Q.; Li, J.; Liu, X.-J.; **Chen, B.**; Leow, W. R.; Wu, R.; Wang, X.; Li, F.; Lum, Y.; Wicks, J.; O'Brien, C. P.; Peng, T.; Ip, A. H.; Sham, T.-K.; Yu, S.-H.; Sinton, D.; Sargent, E. H. Dopant-Tuned Stabilization of Intermediates Promotes Electrosynthesis of Valuable C3 Products. Nat. Commun. 2019, 10 (1), 4807.

33. Proppe, A. H.; Wei, M.; **Chen, B.**; Quintero-Bermudez, R.; Kelley, S. O.; Sargent, E. H. Photochemically Cross-Linked Quantum Well Ligands for 2D/3D Perovskite Photovoltaics with Improved Photovoltage and Stability. J. Am. Chem. Soc. 2019, 141 (36), 14180–14189.

32. Wang, X.; Wu, K.; Blei, M.; Wang, Y.; Pan, L.; Zhao, K.; Shan, C.; Lei, M.; Cui, Y.; **Chen, B.**; Wright, D.; Hu, W.; Tongay, S.; Wei, Z. Highly Polarized Photoelectrical Response in vdW ZrS 3 Nanoribbons. Adv. Electron. Mater. 2019, 5 (7), 1900419.

31. Li, Y. C.; Wang, Z.; Yuan, T.; Nam, D.-H.; Luo, M.; Wicks, J.; **Chen, B.**; Li, J.; Li, F.; de Arquer, F. P. G.; Wang, Y.; Dinh, C.-T.; Voznyy, O.; Sinton, D.; Sargent, E. H. Binding Site Diversity Promotes CO2 Electroreduction to Ethanol. J. Am. Chem. Soc. 2019, 141 (21), 8584–8591.

30. Gao, Y.; Walters, G.; Qin, Y.; **Chen, B.**; Min, Y.; Seifitokaldani, A.; Sun, B.; Todorovic, P.; Saidaminov, M. I.; Lough, A.; Tongay, S.; Hoogland, S.; Sargent, E. H. Electro-Optic Modulation in Hybrid Metal Halide Perovskites. Adv. Mater. 2019, 31 (16), e1808336.

29. Manekkathodi, A.; **Chen, B.**; Kim, J.; Baek, S.-W.; Scheffel, B.; Hou, Y.; Ouellette, O.; Saidaminov, M. I.; Voznyy, O.; Madhavan, V. E.; Belaidi, A.; Ashhab, S.; Sargent, E. Solution-Processed Perovskite-Colloidal Quantum Dot Tandem Solar Cells for Photon Collection beyond 1000 Nm. J. Mater. Chem. A Mater. Energy Sustain. 2019, 7 (45), 26020–26028.

28. Shen, Y.; Shan, B.; Cai, H.; Qin, Y.; Agarwal, A.; Trivedi, D. B.; **Chen, B.**; Liu, L.; Zhuang, H.; Mu, B.; Tongay, S. Ultimate Control over Hydrogen Bond Formation and Reaction Rates for Scalable Synthesis of Highly Crystalline vdW MOF Nanosheets with Large Aspect Ratio. Adv. Mater. 2018, 30 (52), e1802497.

27. Yang, S.; **Chen, B.**; Qin, Y.; Zhou, Y.; Liu, L.; Durso, M.; Zhuang, H.; Shen, Y.; Tongay, S. Highly Crystalline Synthesis of Tellurene Sheets on Two-Dimensional Surfaces: Control over Helical Chain Direction of Tellurene. Phys. Rev. Mater. 2018, 2 (10), 104002.

26. Barbone, M.; Montblanch, A. R.-P.; Kara, D. M.; Palacios-Berraquero, C.; Cadore, A. R.; De Fazio, D.; Pingault, B.; Mostaani, E.; Li, H.; **Chen, B.**; Watanabe, K.; Taniguchi, T.; Tongay, S.; Wang, G.; Ferrari, A. C.; Atatüre, M. Charge-Tuneable Biexciton Complexes in Monolayer WSe2. Nat. Commun. 2018, 9 (1), 3721.

25. Agarwal, A., School for Engineering of Matter, Transport and Energy, Arizona State University, Tempe, Arizona 85287, USA. sefaattin. tongay@asu. edu; Qin, Y.; **Chen, B.**; Blei, M.; Wu, K.; Liu, L.; Shen, Y.; Wright, D.; Green, M. D.; Zhuang, H.; Tongay, S. Anomalous Isoelectronic Chalcogen Rejection in 2D Anisotropic vdW TiS3(1-x)Se3x Trichalcogenides. Nanoscale 2018, 10 (33), 15654–15660.

24. Cai, H.; **Chen, B.**; Blei, M.; Chang, S. L. Y.; Wu, K.; Zhuang, H.; Tongay, S. Abnormal Band Bowing Effects in Phase Instability Crossover Region of GaSe1-xTe x Nanomaterials. Nat. Commun. 2018, 9 (1), 1927.

23. Yang, S.; Qin, Y.; **Chen, B.**; Özçelik, V. O.; White, C. E.; Shen, Y.; Yang, S.; Tongay, S. Novel Surface Molecular Functionalization Route to Enhance Environmental Stability of Tellurium-Containing 2D Layers. ACS Appl. Mater. Interfaces 2017, 9 (51), 44625–44631.

22. Wu, K.; **Chen, B.**; Cai, H.; Blei, M.; Bennett, J.; Yang, S.; Wright, D.; Shen, Y.; Tongay, S. Unusual Pressure Response of Vibrational Modes in Anisotropic TaS3. J. Phys. Chem. C Nanomater. Interfaces 2017, 121 (50), 28187–28193.

21. **Chen, B.**; Wu, K.; Suslu, A.; Yang, S.; Cai, H.; Yano, A.; Soignard, E.; Aoki, T.; March, K.; Shen, Y.; Tongay, S. Controlling Structural Anisotropy of Anisotropic 2D Layers in Pseudo-1D/2D Material Heterojunctions. Adv. Mater. 2017, 29 (34), 1701201.

20. Yang, S.; Cai, H.; **Chen, B.**; Ko, C.; Özçelik, V. O.; Ogletree, D. F.; White, C. E.; Shen, Y.; Tongay, S. Environmental Stability of 2D Anisotropic Tellurium Containing Nanomaterials: Anisotropic to Isotropic Transition. Nanoscale 2017, 9 (34), 12288–12294.

19. Kim, J.; Jin, C.; **Chen, B.**; Cai, H.; Zhao, T.; Lee, P.; Kahn, S.; Watanabe, K.; Taniguchi, T.; Tongay, S.; Crommie, M. F.; Wang, F. Observation of Ultralong Valley Lifetime in WSe2/MoS2 Heterostructures. Sci. Adv. 2017, 3 (7), e1700518.

18. Jin, C.; Kim, J.; Wu, K.; **Chen, B.**; Barnard, E. S.; Suh, J.; Shi, Z.; Drapcho, S. G.; Wu, J.; Schuck, P. J.; Tongay, S.; Wang, F. On Optical Dipole Moment and Radiative Recombination Lifetime of Excitons in WSe 2. Adv. Funct. Mater. 2017, 27 (19), 1601741.

17. Kong, W.; Bacaksiz, C.; **Chen, B.**; Wu, K.; Blei, M.; Fan, X.; Shen, Y.; Sahin, H.; Wright, D.; Narang, D. S.; Tongay, S. Angle Resolved Vibrational Properties of Anisotropic Transition Metal Trichalcogenide Nanosheets. Nanoscale 2017, 9 (12), 4175–4182.

16. Jin, C.; Kim, J.; Suh, J.; Shi, Z.; **Chen, B.**; Fan, X.; Kam, M.; Watanabe, K.; Taniguchi, T.; Tongay, S.; Zettl, A.; Wu, J.; Wang, F. Interlayer Electron–Phonon Coupling in WSe2/hBN Heterostructures. Nat. Phys. 2017, 13 (2), 127–131.

15. Cai, H.; **Chen, B.**; Wang, G.; Soignard, E.; Khosravi, A.; Manca, M.; Marie, X.; Chang, S. L. Y.; Urbaszek, B.; Tongay, S. Synthesis of Highly Anisotropic Semiconducting GaTe Nanomaterials and Emerging Properties Enabled by Epitaxy. Adv. Mater. 2017, 29 (8), 1605551.

14. Wang, C.; Yang, S.; Xiong, W.; Xia, C.; Cai, H.; **Chen, B.**; Wang, X.; Zhang, X.; Wei, Z.; Tongay, S.; Li, J.; Liu, Q. Gate-Tunable Diode-like Current Rectification and Ambipolar Transport in Multilayer van Der Waals ReSe2/WS2 p-n Heterojunctions. Phys. Chem. Chem. Phys. 2016, 18 (40), 27750–27753.

13. Song, Z.; Lv, C.; Liang, M.; Sanphuang, V.; Wu, K.; **Chen, B.**; Zhao, Z.; Bai, J.; Wang, X.; Volakis, J. L.; Wang, L.; He, X.; Yao, Y.; Tongay, S.; Jiang, H. Microscale Silicon Origami. Small 2016, 12 (39), 5401–5406.

12. Wu, K.; Torun, E.; Sahin, H.; **Chen, B.**; Fan, X.; Pant, A.; Parsons Wright, D.; Aoki, T.; Peeters, F. M.; Soignard, E.; Tongay, S. Unusual Lattice Vibration Characteristics in Whiskers of the Pseudo-One-Dimensional Titanium Trisulfide TiS3. Nat. Commun. 2016, 7 (1), 12952.

11. Pant, A.; Torun, E.; **Chen, B.**; Bhat, S.; Fan, X.; Wu, K.; Wright, D. P.; Peeters, F. M.; Soignard, E.; Sahin, H.; Tongay, S. Strong Dichroic Emission in the Pseudo One Dimensional Material ZrS3. Nanoscale 2016, 8 (36), 16259–16265.

10. Wu, K.; **Chen, B.**; Yang, S.; Wang, G.; Kong, W.; Cai, H.; Aoki, T.; Soignard, E.; Marie, X.; Yano, A.; Suslu, A.; Urbaszek, B.; Tongay, S. Domain Architectures and Grain Boundaries in

Chemical Vapor Deposited Highly Anisotropic ReS2 Monolayer Films. Nano Lett. 2016, 16 (9), 5888–5894.

9. Cai, H.; Soignard, E.; Ataca, C.; **Chen, B.**; Ko, C.; Aoki, T.; Pant, A.; Meng, X.; Yang, S.; Grossman, J.; Ogletree, F. D.; Tongay, S. Band Engineering by Controlling vdW Epitaxy Growth Mode in 2D Gallium Chalcogenides. Adv. Mater. 2016, 28 (34), 7375–7382.

8. Kopaczek, J.; Polak, M. P.; Scharoch, P.; Wu, K.; **Chen, B.**; Tongay, S.; Kudrawiec, R. Direct Optical Transitions at K-and H-Point of Brillouin Zone in Bulk MoS2, MoSe2, WS2, and WSe2. Journal of Applied Physics 2016, 119 (23).

7. Wang, C.; Yang, S.; Cai, H.; Ataca, C.; Chen, H.; Zhang, X.; Xu, J.; **Chen, B.**; Wu, K.; Zhang, H.; Liu, L.; Li, J.; Grossman, J. C.; Tongay, S.; Liu, Q. Enhancing Light Emission Efficiency without Color Change in Post-Transition Metal Chalcogenides. Nanoscale 2016, 8 (11), 5820–5825.

6. Cai, H.; Kang, J.; Sahin, H.; **Chen, B.**; Suslu, A.; Wu, K.; Peeters, F.; Meng, X.; Tongay, S. Exciton Pumping across Type-I Gallium Chalcogenide Heterojunctions. Nanotechnology 2016, 27 (6), 065203.

5. Suslu, A.; Wu, K.; Sahin, H.; **Chen, B.**; Yang, S.; Cai, H.; Aoki, T.; Horzum, S.; Kang, J.; Peeters, F. M.; Tongay, S. Unusual Dimensionality Effects and Surface Charge Density in 2D Mg(OH)2. Sci. Rep. 2016, 6 (1), 20525.

4. Wang, G.; Robert, C.; Suslu, A.; **Chen, B.**; Yang, S.; Alamdari, S.; Gerber, I. C.; Amand, T.; Marie, X.; Tongay, S.; Urbaszek, B. Spin-Orbit Engineering in Transition Metal Dichalcogenide Alloy Monolayers. Nat. Commun. 2015, 6 (1), 10110.

3. Meng, X.; Pant, A.; Cai, H.; Kang, J.; Sahin, H.; **Chen, B.**; Wu, K.; Yang, S.; Suslu, A.; Peeters, F. M.; Tongay, S. Engineering Excitonic Dynamics and Environmental Stability of Post-Transition Metal Chalcogenides by Pyridine Functionalization Technique. Nanoscale 2015, 7 (40), 17109–17115.

2. Aierken, Y.; Sahin, H.; Iyikanat, F.; Horzum, S.; Suslu, A.; **Chen, B.**; Senger, R. T.; Tongay, S.; Peeters, F. M. Portlandite Crystal: Bulk, Bilayer, and Monolayer Structures. Phys. Rev. B Condens. Matter Mater. Phys. 2015, 91 (24), 245413.

1. **Chen, B.**; Sahin, H.; Suslu, A.; Ding, L.; Bertoni, M. I.; Peeters, F. M.; Tongay, S. Environmental Changes in MoTe2 Excitonic Dynamics by Defects-Activated Molecular Interaction. ACS Nano 2015, 9 (5), 5326–5332.